

Zoning in the White Pine Copper Deposit, Ontonagon County, Michigan

ALEXANDER C. BROWN

Abstract

The top of the cupriferous zone in the lower Nonesuch Shale of Northern Michigan is characterized by an abrupt zonation of Cu-Fe sulfides which represents a chemical and mineralogic transition into the overlying barren pyritic shale. This narrow fringe forms a blanket-like surface which cross-cuts bedding at shallow angles. Although the ore horizons show strict stratigraphic controls, the actual upper limit of copper mineralization in the White Pine area occurs at various stratigraphic levels 20 to 50 feet above the base of the Nonesuch. A strong inverse correlation exists between the height of the fringe surface in the Nonesuch Shale and the amount of copper in the lowermost shale unit. In local areas along fracture zones, a very minor amount of copper has been channeled upward along some fracture zones producing local anomalous elevations in the fringe surface. The White Pine fault is clearly a post-ore structure which offsets the fringe surface.

The ore minerals of the Cu-Fe transition show a step-by-step replacement of iron-rich sulfides by paragenetically later copper-rich sulfides. The sequence in ascending order is chalcocite, djurleite, digenite, bornite, chalcopyrite and pyrite. Abnormal concentrations of Cd and Pb occur immediately above the cupriferous zone.

These observations suggest, as others have proposed, that the mineralization of the basal Nonesuch was accomplished by the introduction of copper from the underlying Copper Harbor Conglomerate soon after deposition of the Nonesuch Shale. Copper deposition in the sulfur-rich reducing environment of the Nonesuch occurred in large part by replacement of original pyrite along an upward advancing mineralization front (fringe). Metals more soluble than copper were apparently swept upward before the copper front.

Simple mathematical models of infiltration and diffusion processes were designed to test the above hypothesis. Each model results in a mineralization period of about 10^6 years. It seems entirely possible that these mechanisms could account for the transport of copper into the early unconsolidated Nonesuch sediment. Salt sieving and gravity stratification phenomena may have assisted these basic processes. The ultimate source of copper is open to speculation, but may lie in primary magmatic contributions or extraction of metals from the thick volcanic-sedimentary pile of the Keweenaw basin.

Introduction

THE famous stratiform copper deposit at White Pine, Michigan has often been cited as a typical example of a syngenetic base metal deposit. The principal observations supporting this interpretation are the vast lateral extent of mineralization, the strict confinement of ore to certain well-defined stratigraphic horizons, and the uniformity of tenor along bedding. Recent studies suggest instead that the mineralization took place after deposition of the lower Nonesuch Shale and before moderate regional deformation of this late Keweenaw formation.

This paper presents the writer's contributions to this continuing debate. Conclusions are drawn principally from (1) the configuration of the mineralized zone, (2) mineralogic zoning, and (3) preliminary models of the proposed mineralization process. Particular emphasis is given to the genetic significance of

an abrupt chemical discontinuity which occurs at the upper limit of copper mineralization.

History of Related Investigations

The present investigation was prompted by the reports of White and Wright (1954) and White (1960) which noted the occurrence of a narrow Cu-Fe transition (fringe) between the cupriferous zone and the overlying barren pyritic shales. The stratigraphic position of this fringe feature was shown to vary between vertical drill core sections which transect the lower Nonesuch, and in many areas the cupriferous zone was found to include so-called "barren" beds above the usual ore horizons. White (1960) described the fringe as consisting of narrow mineralogic subzones with the sequence chalcocite → bornite → chalcopyrite → pyrite in ascending order. Traces of greenockite (CdS) were observed in the

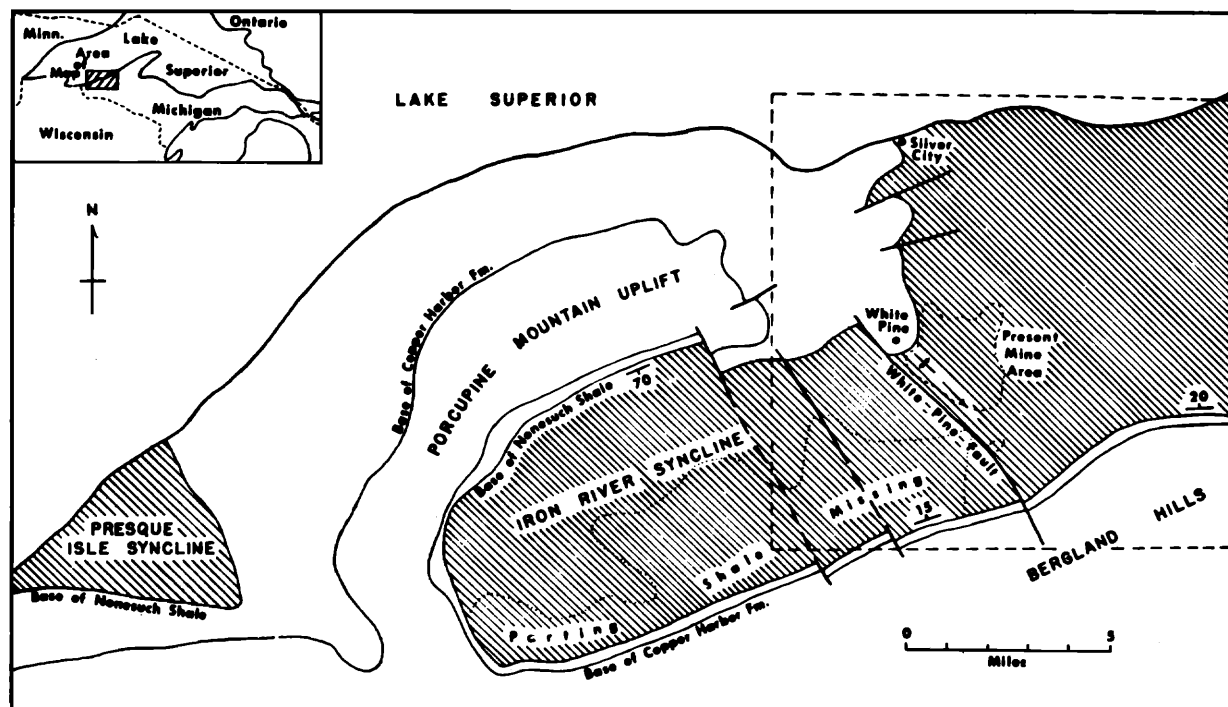


FIG. 1. Index map showing the major structural and stratigraphic features of the White Pine and surrounding regions. Area of present study is delineated by dashed lines. (Modified after White and Wright, 1966).

pyritic shales immediately overlying the cupriferous zone. These observations led White to suggest that the Cu-Fe transition represents the final position of a copper mineralization front which advanced upward into initially pyritic shales.

Kelly (1962) showed from examinations of drill cores and drill log data that the stratigraphic position of the Cu-Fe transition varies as much as 30 feet across the White Pine area. This survey, together with communications with White and Wright and C. O. Ensign, Jr. were primarily responsible for outlining the investigations upon which this report is based. Studies began in the fall of 1964 and continued until 1968. Interim reports were presented by Brown (1965), Brown and Trammell (1966),

and Brown (1968). During these investigations, White and Wright (1966) published their observations on the fringe configuration and sulfide zoning from studies of a broad region including both the Iron River and Presque Isle synclines. The writer's examinations were confined to a detailed study in the White Pine area (Fig. 1). A close agreement exists between the findings of White and Wright and those of the writer.

Other investigations to which the present work is closely related include those of Wiese (1960), Carpenter (1963), Chaudhuri and Faure (1967), and Hamilton (1967).

Geologic Setting

General

The geology of the White Pine district is thoroughly described in the classic papers of Butler and Burbank (1929) and White and Wright (1954). Recently Ensign et al. (1968) have given a more up-to-date comprehensive review. Only a very brief description of the most pertinent features is given here.

Table 1 shows that the 600–720 foot section of the Nonesuch Shale at White Pine lies between two thick Upper Keweenawan red-bed formations known as the Copper Harbor Conglomerate (below) and the Freda Sandstone (above). Beneath the Copper Har-

Table 1. Regional stratigraphic section of Keweenawan rocks in the White Pine area (data from Ensign et al., 1968)

Age	Formation	Thickness (ft.)
Upper Keweenawan	Freda Sandstone	12,000 ±
	Nonesuch Shale	500–720
	Copper Harbor Conglomerate	350–6000
Middle Keweenawan	Portage Lake Lava Series	30,000 ?

bor formation is the Portage Lake Lava Series which is well-known for its native copper deposits some 20–80 miles northeast of White Pine.

The copper mineralization at White Pine is represented principally by disseminations of chalcocite and native copper in the basal portions of the Nonesuch Shale and in the uppermost chloritic portions of the Copper Harbor formation. Detailed descriptions of this stratigraphic section have been given in several recent publications including White and Wright (1954, 1966), Wiese (1960) and Hamilton (1967). Figure 2 and Table 2 show the stratigraphy and nomenclature of units referred to in this paper.

Structure

The White Pine area is located on the northeast-southwest trending axis of the Iron River syncline between the Porcupine Mountain uplift on the north and the Bergland Hills to the south (Fig. 1). Dips are gentle (15–30°) for the most part, but are steep

Table 2. Stratigraphic section at White Pine

Bed No.	Bed Name (underlined) and Description
65	<u>Marker</u> : thinly laminated siltstone and shale with tiny white calcite blebs
63	Black thinly laminated siltstone
61	<u>Stripey</u> : thinly laminated light-grey siltstone and shale
50	<u>Tiebel</u> sandstone and green massive siltstone
49	<u>Red and Grey</u> laminated siltstone and shale
47	<u>Widely</u> laminated shales and siltstone
46	<u>Upper Zone of Values</u> : dark grey massive siltstone
44	<u>Grey Massive</u> : grey-brown massive siltstone
43	<u>Thinly</u> laminated sandstone, siltstone and black shales
41	<u>Upper Transition</u> : grey sandstone with black shale partings
30	<u>Upper Sandstone</u> : red and grey sandstone with interbedded siltstone and shales
29	<u>Tiger</u> : red and grey laminated siltstone and shales
27	<u>Top Zone</u> : widely laminated shales and siltstone
26	<u>Dark Grey Massive</u> siltstone
24	<u>Brown Massive</u> : brown-red massive siltstone
23	<u>Domino</u> : thinly laminated sandstone, siltstone, and black shale
21	<u>Lower Transition</u> : grey sandstone with black shale partings
10	<u>Lower Sandstone</u> : red and grey sandstone

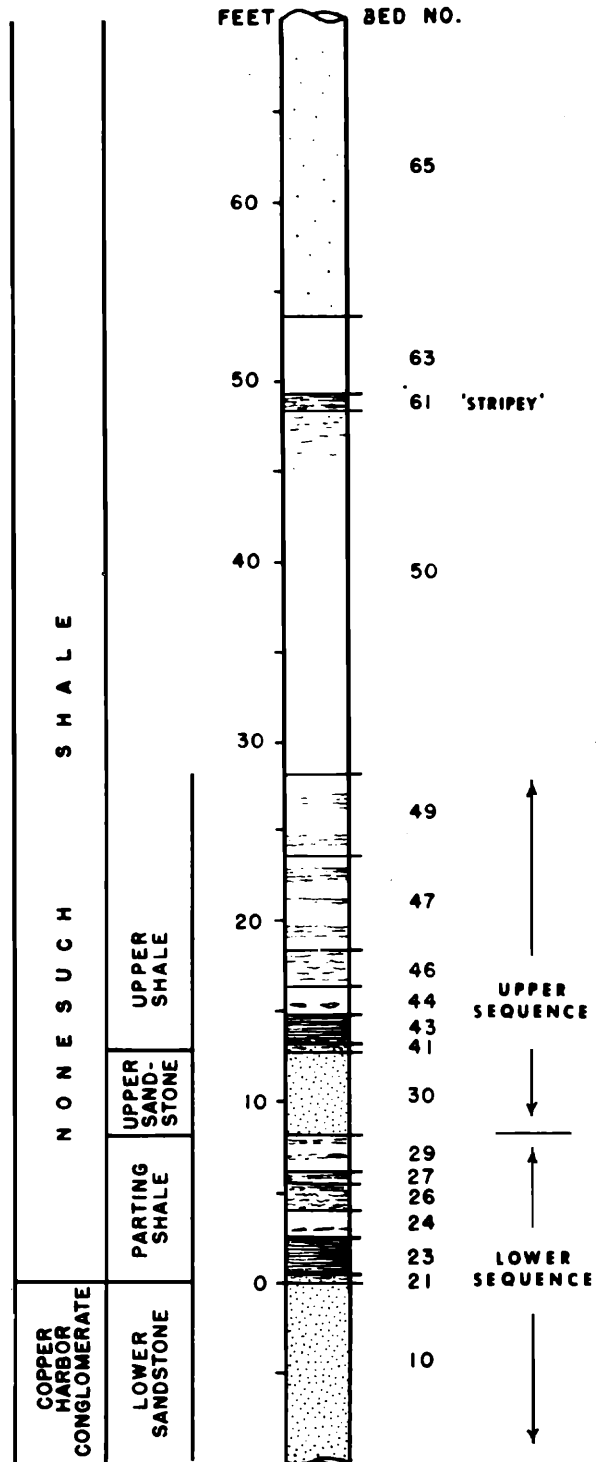


FIG. 2. Stratigraphic column of basal Nonesuch and uppermost Copper Harbor formations (see description in Table 2).

and locally overturned where the Nonesuch lies against the south limb of the Porcupine Mountain uplift.

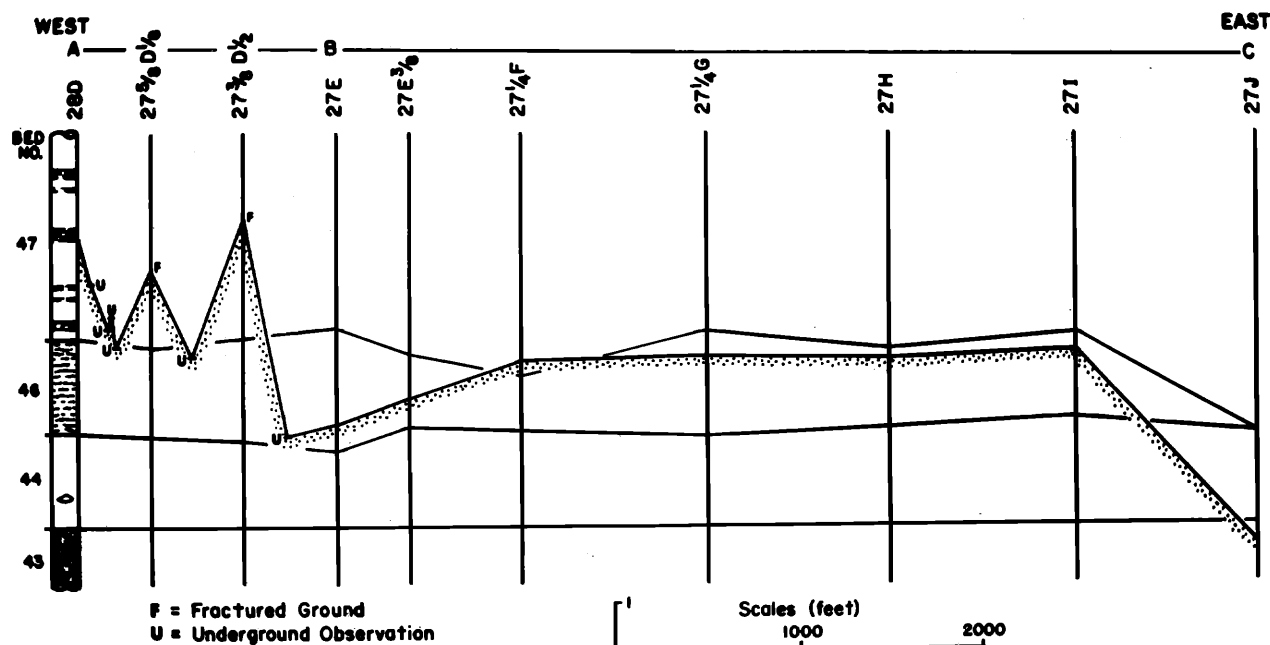


FIG. 3. Profile showing the position of the Cu-Fe transition fringe along an east-west section (for positions ABC, see inset of Fig. 5).

The structure of the immediate White Pine area is dominated by the northwest-trending White Pine fault which occurs along the western margin of the present mine area. This is a reverse right-hand fault with a steep dip to the northeast. Southwest of the mine, the horizontal displacement is estimated at 3,000–5,000 feet and the vertical component is estimated to be 1,500 feet. Southeast of the mine, the vertical component seems to be about 240 feet. Other faults of comparable magnitude are known to the east and west of the White Pine area.

Structures on either side of the White Pine fault are under continual review by the mine personnel. Underground exposures and drill core studies have revealed many faults of small vertical displacement over a wide area encompassing most if not all of the area of this study. Associated with these minor faults are numerous subsidiary fracture systems. Some of the smaller faults and fractures may have developed as a result of differential compaction of the Keweenaw sediments. Others may have originated at a later time during the period of regional deformation responsible for such major structures as the Porcupine Mountain uplift and the White Pine fault. A shallow northwest-southeast trending anticline and a high density of minor faults and fractures in the mine area are probably results of drag on the White Pine fault.

Configuration of the Mineralized Zone

Because of some similarities in ore distribution, mineralogy and sulfide zoning, the White Pine copper deposit has been spoken of as the North American analog of the well-known Zambian copper ores. The latter exhibit broad zones of chalcocite, bornite, chalcopyrite and pyrite which are attributed to oscillations in the shoreline of the depositional basin (Garlick, 1961). If the White Pine mineralization had an equivalent syngenetic origin, one might expect to find similar lateral oscillations of sulfide zones within the Nonesuch strata. At least one major offlap-onlap sequence is indicated during lower Nonesuch sedimentation by a slight erosional unconformity at the top of the Parting Shale unit. If, on the other hand, the top of the cupriferous zone at White Pine proved to be a single smooth or gently undulating surface without lateral oscillations within bedding, one would be encouraged to think in terms of mineralization by an upward influx of copper as proposed by White (1960).

In order to test these concepts, the writer undertook two studies of the fringe configuration. The first consisted of detailed examinations of the mineralization in closely spaced drill cores and underground exposures, and equally detailed studies of several regionally spaced drill cores. In the second study, the nature and stratigraphic position of the Cu-Fe transition were noted in all drill core available at the White Pine mine. The purpose of this investi-

gation was to establish the configuration of the fringe surface across the mine area.

Detailed Studies

Method.—Ten drill cores at a spacing of 1,000 feet or less (see east-west profile ABC, Fig. 3) were examined, inch-by-inch, in order to log all disseminated and fracture mineralizations, lithologies and small scale structures. Observations were made on continuous, sawn core surfaces under a low-power (20–40×) binocular microscope. Each core consisted of a 60–70 foot section extending from the No. 10 to the No. 65 bed (i.e., through the entire cupriferous zone and well into the overlying pyritic zone). The uppermost occurrence of bornite was recorded as the top of the cupriferous zone.

Similar examinations were made of more widely spaced drill cores along two north-south trending lines (DBAE and FG, Fig. 4). Five other drill cores were studied in a similar manner in a search for repetitions of the fringe on a regional basis.

In addition, two underground investigations were made. The first was a continuous traverse of mine support pillars between drill holes 27E and 28D (Fig. 3). The second was in the area of the experimental Phase II longwall panel (Fig. 5). In these underground studies, the Cu-Fe transition was identified by its characteristic mineralogy and by use of a copper sensitive molybdate-based paint.

Throughout these studies, the footage of the Cu-Fe transition was recorded with respect to the top of the No. 43 bed, a distinct dark gray marker horizon which is readily identified in all vertical sections through the lower Nonesuch.

Results.—Only one Cu-Fe transition was observed in all vertical drill core and mine sections examined. From this result it is assumed that the fringe forms a single line in two-dimensional profiles as shown in Figures 3 and 4, and a single surface in three dimensions as illustrated in Figure 4. It is important to note, however, that Manfred Jost, University of Mainz, has locally found Cu-Fe sulfides and greenockite in the upper part of the Parting Shale unit beneath the normal fringe position in the Upper Shale section. This feature was not observed by other investigators including the writer, but has been verified in a limited area of the mine. Further work will be required to determine the extent and configuration of this type of occurrence and its bearing on results presented in this paper.

In Figure 3, the Cu-Fe transition varies in stratigraphic position from the No. 43 bed to the No. 47 bed (a vertical thickness of about 7 feet measured with respect to the top of the No. 43 bed) over a lateral distance of about 6,500 feet. Fluctuations in

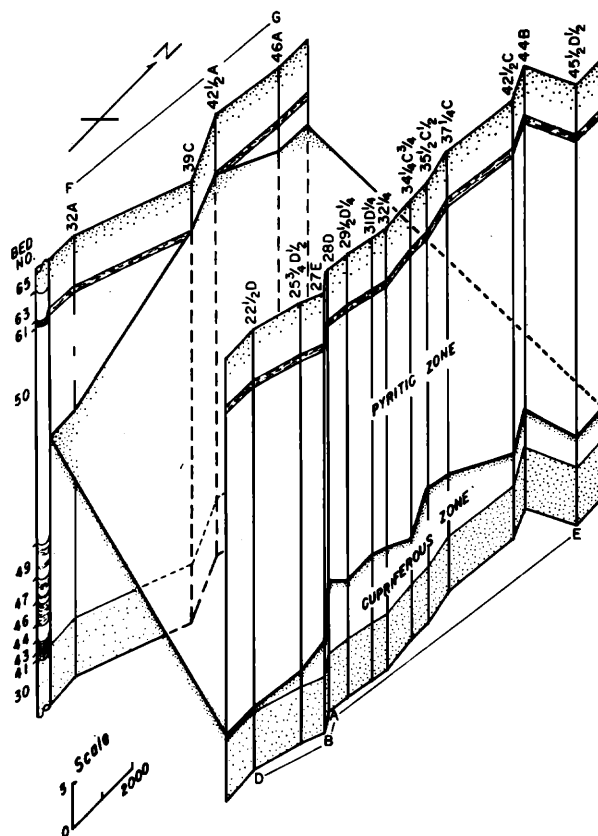


FIG. 4. Position of the Cu-Fe transition in two north-south trending lines of drill cores which show a major transaction of bedding by the fringe surface. Profiles have been displaced laterally in east-west directions for viewing purposes (for actual positions, see lines DBAE and FG in inset of Fig. 5).

the trace of the fringe are most pronounced where the shale has been fractured as in drill cores 27 3/8 D 1/2 and 27 3/8 D 1/2. In underground examinations of nearby unfractured ground (see the notation "U" in Fig. 3), it was found that the fringe occurs at levels which are typical of the regional trend of the fringe surface in this portion of the mine. From these data and similar observations in other areas, it appears that the fringe surface is commonly elevated in the immediate vicinity of some fractures and occurs at lower and more regionally consistent levels a few inches to either side of such fractures or fracture zones.

A pronounced transaction of bedding by the fringe surface is clearly demonstrated between the two closely spaced north-south trending lines of drill holes (Fig. 4). Whereas the fringe occurs at the No. 43–47 level in the easterly profile, it is found at the No. 50–61 level in the second line of drill holes located some 1,000–2,000 feet further west. About 30 feet of shale is transected by the fringe surface between these two profiles.

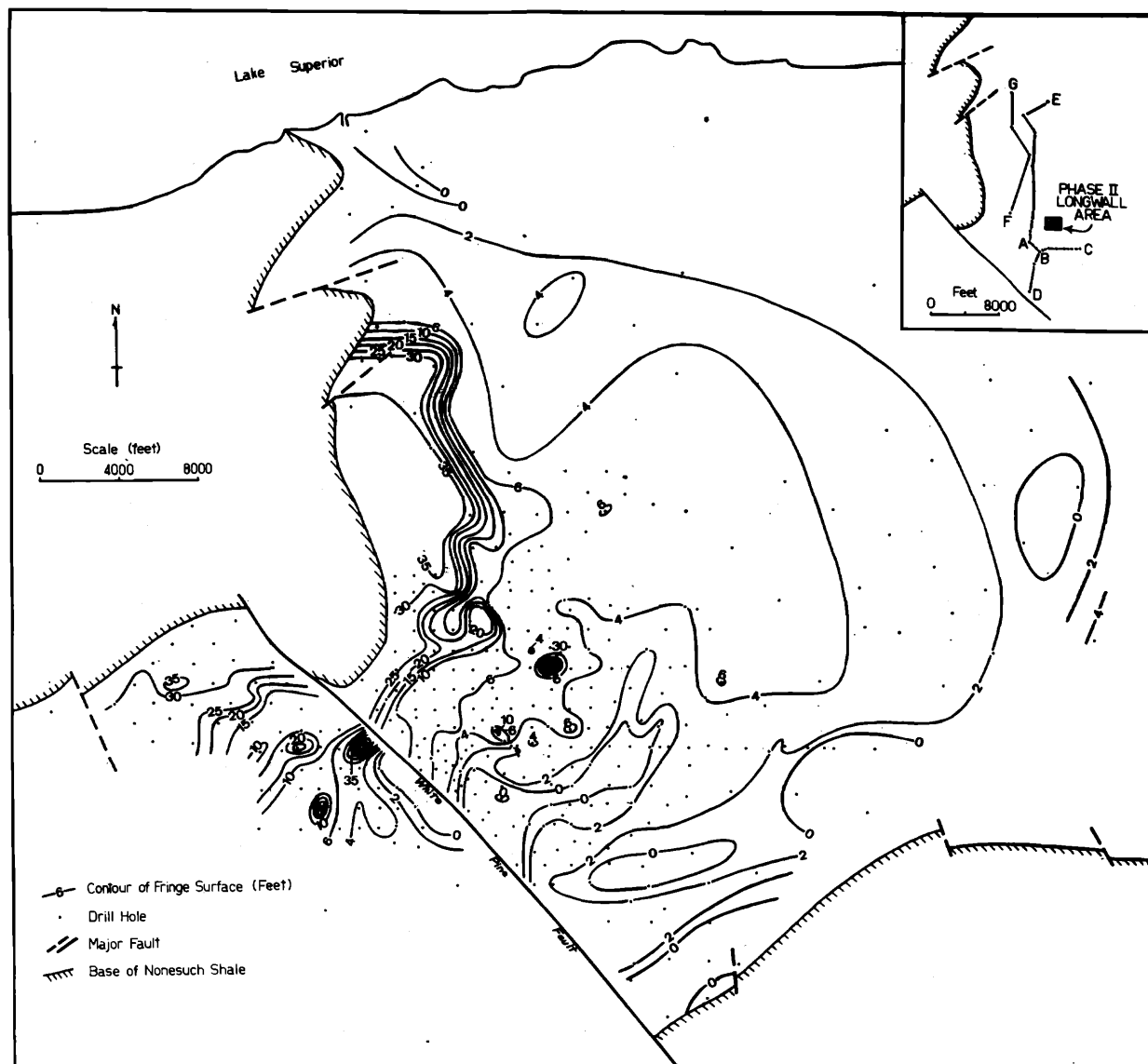


FIG. 5. Contour map of the fringe surface in the White Pine area. Top of No. 43 bed represents the zero datum.

Regional Studies

Method.—Detailed microscopic studies, which agree with the observations of White and Wright, lead to the conclusion that the Cu-Fe transition forms a single blanket-like surface without lateral oscillations. With this picture in mind, a survey was then made of all available drill core (311 in all) to determine the regional configuration of this undulating surface across the White Pine area. Binocular examinations were generally restricted to identification of the Cu-Fe transition, with notations of its footage and stratigraphic position, its mineralization and any apparently related structural or sedimentary features. Occasional scans were made of complete core sections from the No. 10 to the No. 65 bed as

a check for possible repetitions of the fringe; none were found.

Results.—The regional configuration of the Cu-Fe transition relative to the top of the No. 43 bed is illustrated by the contoured surface in Figure 5. Over a very large area in the south, southeast and east, the fringe is positioned at or near the top of the No. 43 horizon. Toward the north, northwest and west, there is a somewhat smaller area in which the Cu-Fe transition occurs about 30 feet higher in the Nonesuch strata, at or near the No. 61 bed. In brief, there are two major plateaus in the fringe surface with a relatively distinct steep slope joining these two levels. Both plateaus coincide closely with well-mineralized strata: the lower plateau at the

Table 3. Linear correlation coefficients for the relations between thickness, assay and thickness times assay and height of the Cu-Fe transition. N = number of drill holes included in the regression analysis, R = product moment correlation coefficient, C = 95% confidence interval on R. Where N is less than 11 in (a) or less than 36 in (b), data were not available for some drill cores.

(a) Data from diamond drill holes on west side of White Pine fault									
Beds	Thickness versus Height			Assay versus Height			Thickness x Assay versus Height		
	N	R	C	N	R	C	N	R	C
21+23	11	-.49	.16 to -.84	11	-.83	-.47 to -.96	11	-.92	-.73 to -.98
26	11	.37	.79 to -.30	11	.34	.78 to -.32	11	.44	.82 to -.22
41+43	11	.81	.95 to .40	11	-.54	.09 to -.86	11	-.19	.46 to -.71
46	11	.52	.87 to -.16	11	-.40	.31 to -.82	10	.00	.63 to -.63

(b) Data from diamond drill holes of the White Pine area located on both sides of the White Pine fault									
Beds	Thickness versus Height			Assay versus Height			Thickness x Assay versus Height		
	N	R	C	N	R	C	N	R	C
21+23	21	-.34	.11 to -.67	21	-.79	-.55 to -.91	36	-.88	-.77 to -.94
26	21	.37	.69 to -.07	21	.08	.50 to -.36	21	.32	.66 to -.13
41+43	21	.56	.80 to .17	21	-.61	-.24 to -.82	21	-.32	.13 to -.66
46	14	.38	.76 to -.19	14	-.31	.27 to -.72	14	-.03	.51 to -.55

No. 41–47 level, and the upper plateau at the upper No. 50–65 level. The steep slope between the plateaus cross-cuts a relatively thick and poorly-mineralized section consisting of the No. 49 and the lower No. 50 beds.

White and Wright (1966) observed that an inverse relationship apparently exists between the thickness of the cupriferous zone and the thickness of the basal bed of the Parting Shale (No. 23 bed). They suggested that this correlation could be explained if copper entered the shale from below in rather uniform amounts across the base of the Nonesuch. Where the normally well-mineralized No. 23 bed is thick, a large portion of the incoming copper would be deposited at the immediate base of the Nonesuch formation, and proportionately less copper would remain for mineralization of higher beds. Where the No. 23 bed is thin, the mineralization front would advance to a much higher position in the Nonesuch section because only a small amount of the total introduced copper would be deposited in the basal shale unit.

Modifying White and Wright's (1966) model somewhat, the ultimate height of the fringe would seem to depend not only on the thickness of well-mineralized beds but also on the *amount* of mineralization in each bed. In fact, the rate of advance of the mineralization front through any bed would probably have varied inversely with the product of its thickness and assay values (i.e., the greater the total copper in any bed, the longer the residence time of the front in that bed during mineralization).

The assumption that complete mineralization took place along a sharp mineralization front is supported by the observed abruptness of the Cu-Fe transition.

This modification of White and Wright's (1966) basic idea has been applied in a quantitative study involving both thickness and assay values of a number of drill cores at White Pine. A very strong inverse correlation was found between the fringe height (relative to the top of the No. 43 bed) and *thickness times assay data* of the No. 21 + 23 beds.¹ This relationship is illustrated for two groups of drill cores in Figure 6 and Table 3. A very strong inverse correlation was found between the fringe height (relative to the top of the No. 43 bed) and *thickness times assay data* of the No. 21 + 23 beds. Rather good inverse correlations exist between the fringe height and *assay* values for the No. 21 + 23 beds, but the correlations for *thickness* data alone seem much poorer. From these statistical results, it seems probable that the No. 21 + 23 beds had a marked control over the ultimate thickness of the mineralized Nonesuch section, and that the specific controlling properties are, in order of decreasing importance, (1) total mineralization (thickness times assay), (2) assay and (3) thickness.

Similar calculations indicate that the No. 26 and No. 46 beds had no particular influence on the height of the fringe. However, thickness and assay proper-

¹ The combined mineralization of the No. 21 and No. 23 beds (No. 21 + 23 beds) has been used rather than the mineralization of either bed alone because the No. 21 bed is transitional into the No. 23 bed and together they form one well-defined stratigraphic unit.

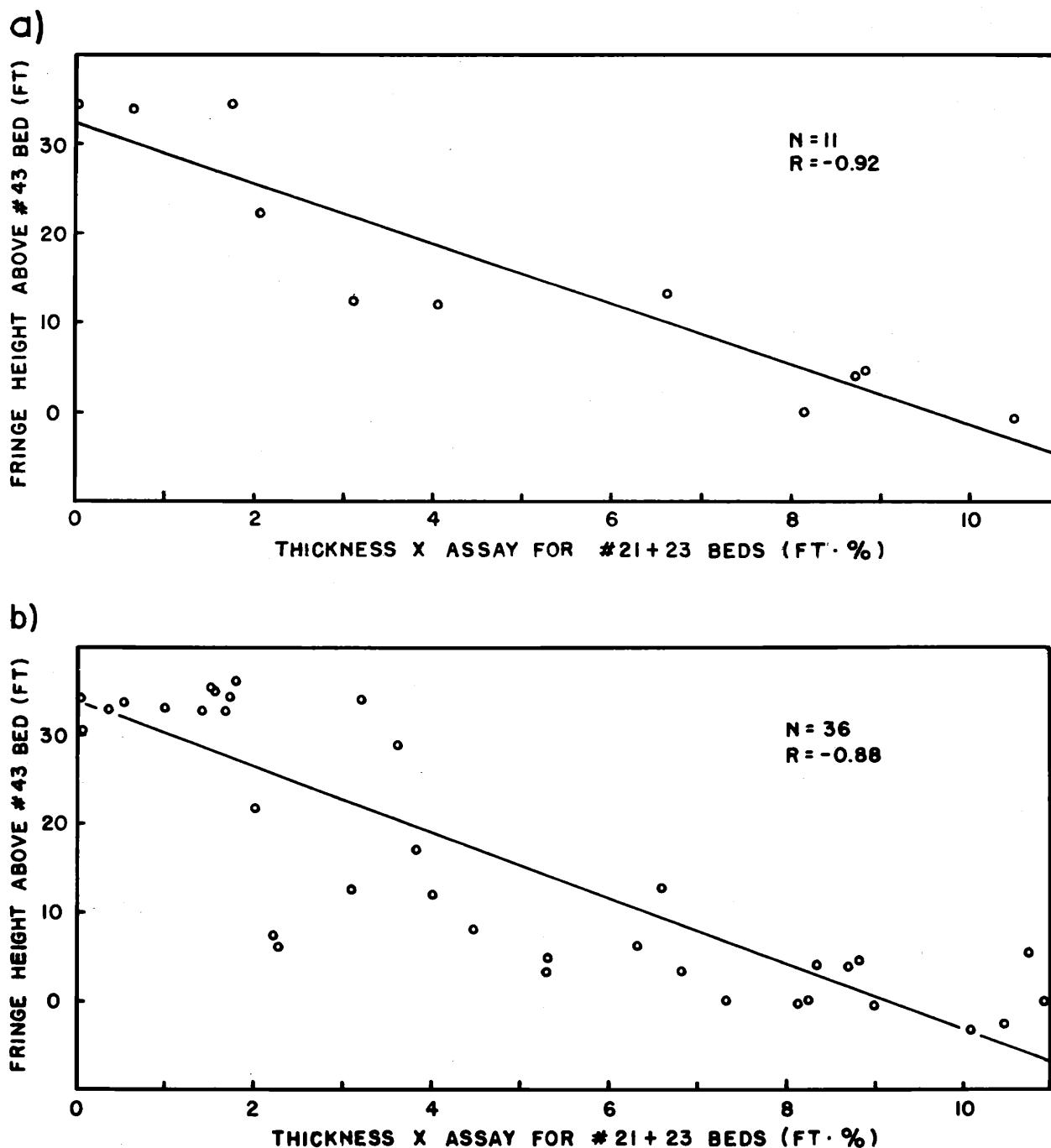


FIG. 6. Plot of the fringe position (relative to the top of the No. 43 bed) versus the thickness $\times$ assay values of the No. 21 + 23 beds.

ties of the No. 41 + 43 beds do appear to have independent correlations with the fringe height. The opposed trends of the positive correlations derived from thickness data and the inverse correlations derived from assay data are probably responsible for the lack of any significant relationship between thickness times assay data and the fringe height for the

No. 41 + 43 beds. This situation is unlike that of the comparable No. 21 + 23 beds of the Parting Shale described earlier.

The strong inverse correlation between the total mineralization of the No. 21 + 23 beds and the position of the Cu-Fe transition has a very significant bearing on genesis of the White Pine mineralization.

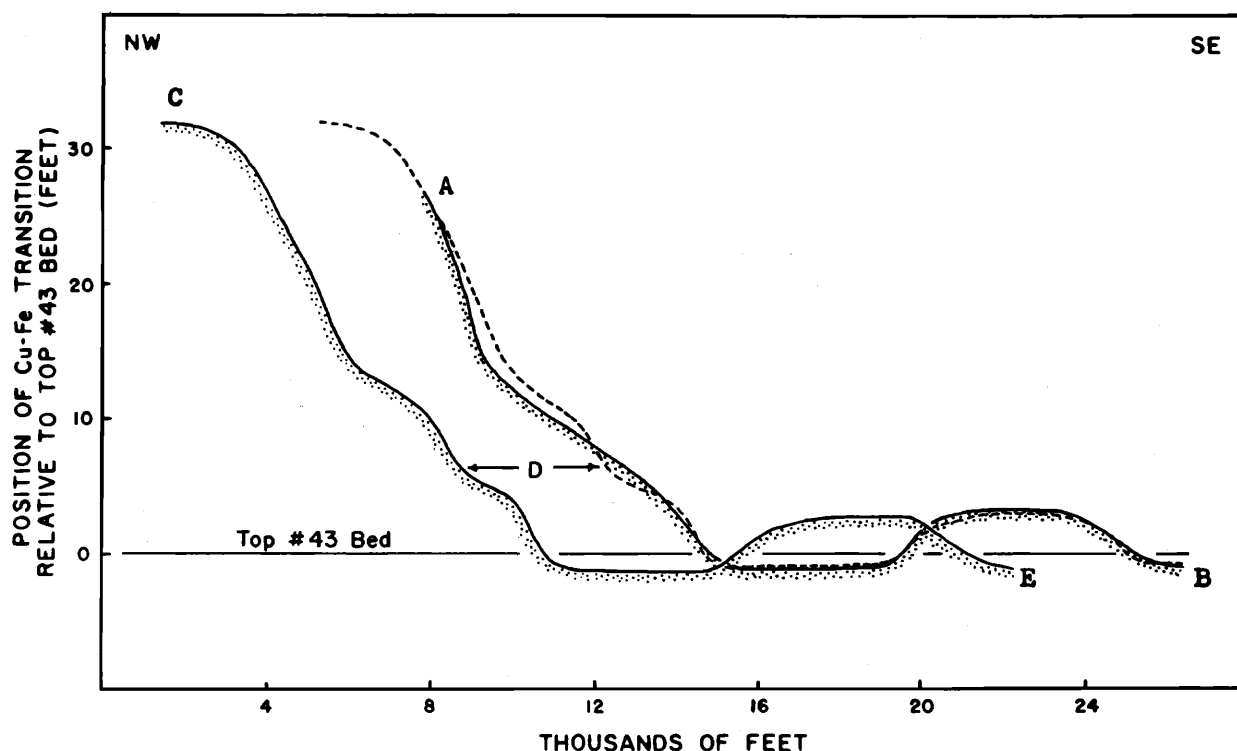


FIG. 7. Traces of the Cu-Fe transition on the east (curve A-B) and west (curve C-E) sides of the White Pine fault. The dashed line is generated by translating curve C-E 3,800 feet southeastward (distance D).

It would be difficult to explain such a relationship between the fringe position and the mineralization of a bed some 20–50 feet below the fringe surface unless the mineralizing solutions did indeed pass through those basal Nonesuch beds. The available evidence thus supports the proposal of White and Wright (1966) that the copper mineralization was accomplished by ascending cupriferous solutions which entered the shale from the underlying Copper Harbor formation. This is the premise basic to the infiltration and diffusion models constructed in a later part of this paper.

The evidence thus far points toward a stratigraphic control of the ultimate position of the mineralization front. A structural influence on the fringe configuration is suggested by the geographic coincidence of the high plateau of the fringe surface with the present Porcupine Mountain uplift, but in reality it is the coincidence of a sandy equivalent of the No. 21 + 23 beds with the Porcupine Mountain area which requires explanation. C. O. Ensign, Jr. (personal communication, 1968) has suggested that a slightly positive, partially submerged portion of the early Nonesuch basin existed in the general area of the present domal structure. A sandy sediment was deposited across this region while a shale (No. 21 + 23 beds) was deposited in deeper portions of the basin. This slightly elevated portion of the

basin floor was apparently buried during early Parting Shale sedimentation and may have been an early manifestation of a later more intense uplift now represented by the Porcupine Mountain structure. This concept permits explanation of the fringe configuration within the stratigraphically-controlled model described above.

The concept of a mineralization front accounts very well for (1) the general coincidence of the fringe with sulfide-rich stratigraphic sections, and (2) the existence of two distinct plateaus in the fringe configuration. The residence time of the front should be long for mineralization of sulfide-rich beds and short for sulfide-poor beds. Hence the probability is high that the fringe would be located in well-mineralized strata at the end of the mineralization period. Plateaus in the fringe surface, notably at the Upper Shale and No. 61 levels, simply represent sulfide-rich stratigraphic sections attained by the front when mineralization ceased.

Structural-related irregularities in the fringe surface in the immediate vicinity of some fractures have been mentioned above. The regional distribution and density of such fractures is not yet clear, but there is no evidence that they exerted any major control over the migration of copper through the lower Nonesuch. Although major fractures capable of guiding significantly large amounts of copper were

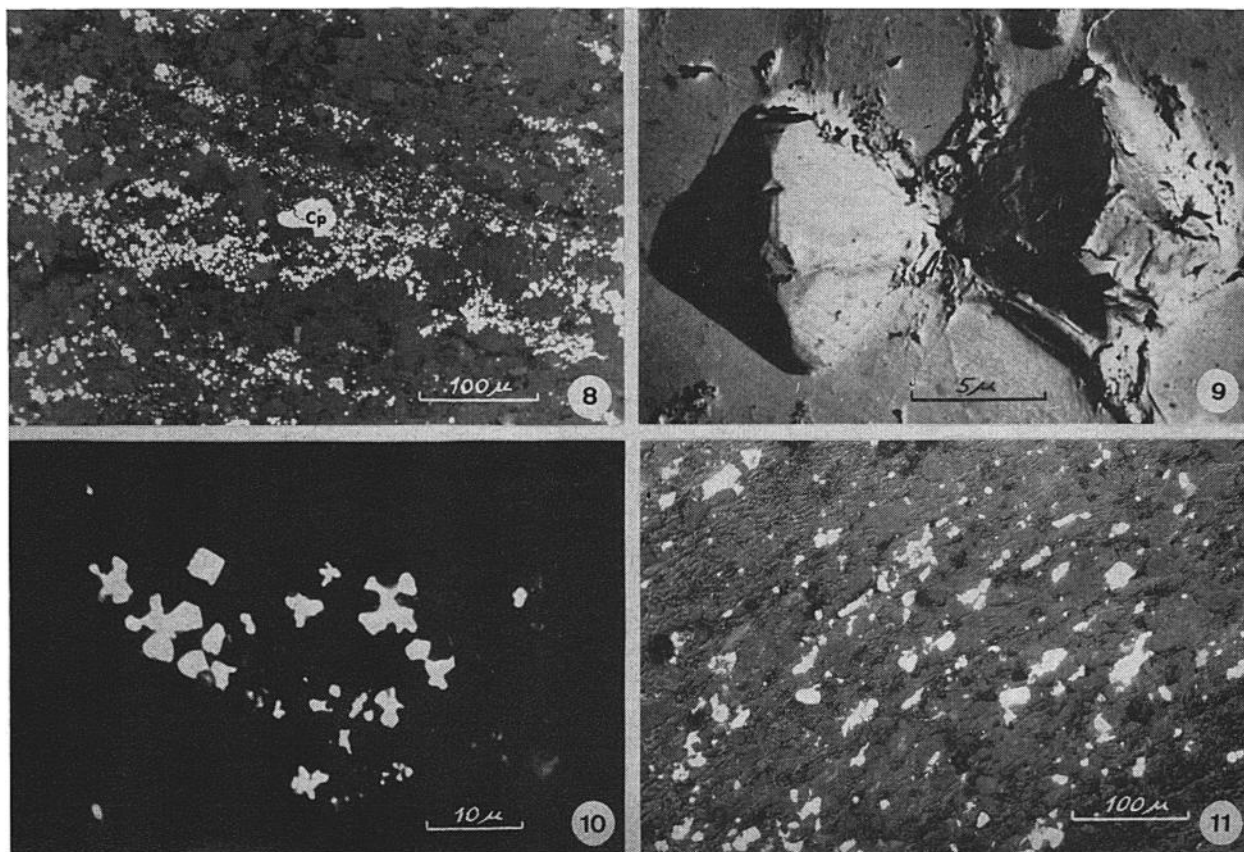


FIG. 8. Disseminated pyrite euhedra in a dominantly quartz-chlorite gangue. A relatively coarse grain of chalcopyrite (Cp) occurs in the center of the field. Reflected light, in air.

FIG. 9. Electron photomicrograph of octohedral pyrite grains (courtesy of I. S. Zajac).

FIG. 10. Corroded grains of pyrite showing typical skeletal structures. Reflected light, in oil.

FIG. 11. Irregular disseminated grains of chalcocite in a dominantly quartz-chlorite-carbonate gangue typical of the cupriferous zone. Reflected light, in air.

probably formed after mineralization was essentially complete, local fracture-related irregularities in the fringe surface testify to a very minor amount of mineralization after some fracturing.² The similarity of the paragenetic trends of sulfides in these fractures and sulfides in the fringe suggest a close genetic link between these two mineralizations. The small amount of copper in fractures and adjacent wall-rock where the fringe is elevated may represent very late stage copper which moved upward into the pyritic zone at a time when the Nonesuch was initially fractured, probably toward the end of the mineralization period.

The fringe surface experienced a major post-ore displacement along the White Pine fault. Traces of the fringe surface projected onto the fault plane (Fig. 7) show a lateral displacement of about 3800

feet, disregarding a small contribution from rotational movement. This estimated offset is intermediate in magnitude between previous estimates of 3,000 feet (Ensign et al. 1968) and one mile (White and Wright, 1954).

Mineralogy

Recent mineralogic studies of the White Pine deposit have concentrated on discrete facies of the lower Nonesuch Shale: Wiese (1960) on its petrology and geochemistry, Carpenter (1963) on late fracture mineralization, White and Wright (1966) on sulfide zoning from regional studies, Hamilton (1967) on native copper occurrences in the uppermost Copper Harbor formation, and Chaudhuri and Faure (1967) on the age of the disseminated mineralization and associated rock units.

The following discussion is concerned chiefly with the mineralogy of the abrupt Cu-Fe transition between the cupriferous and pyritic zones. Thirteen ore minerals have been identified at the position of

² It is important to distinguish between this fracture-controlled mineralization and the still later vein mineralization described by Carpenter (1963). The late vein sulfides are distinguished by a paragenetic trend which is the reverse of that in the fringe.

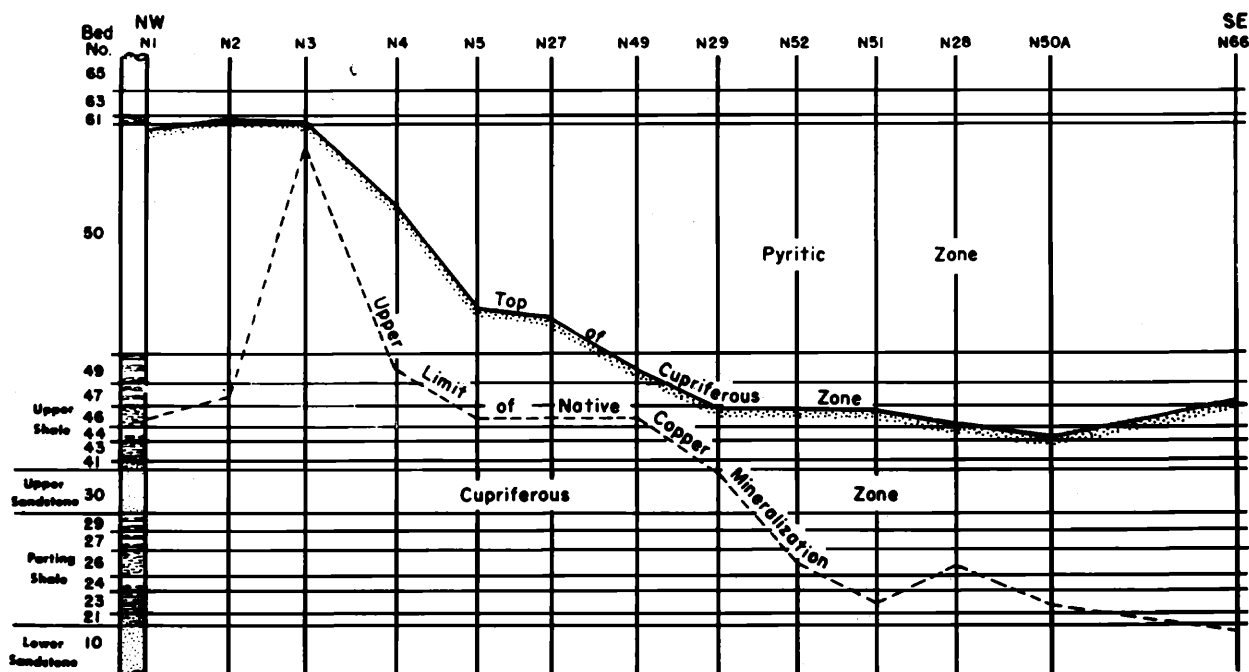


FIG. 12. Sketch showing the close spatial relationship between the stratigraphic position of the Cu-Fe transition and the highest occurrence of native copper in a northwest-southeast trending profile of 13 drill cores located on the west side of the White Pine fault. Horizontal and vertical scale are approximately 1 inch to 2,400 and 12 feet, respectively.

this fringe zone: bornite, chalcocite, chalcopyrite, covellite, digenite, djurleite, galena, greenockite, hematite, marcasite, pyrite, sphalerite, and wurtzite.

Mineralogy of the Pyritic Zone

Pyrite.—Pyrite is the dominant sulfide of the unmineralized Nonesuch Shale, and occurs for the most part as well-disseminated euhedra (Figs. 8 and 9) within chlorite grains. The average grain size is 5–10 μ , although unusually large grains up to 100 μ

in diameter have been observed. Pyrite is also seen as clusters of euhedra in the form of spherical structures which resemble framboidal groups found in many dark grey shales (e.g., the Kupferschiefer and Mount Isa). Pyritic aggregates larger than 1 mm across are typically lenticular with long dimensions parallel to bedding, possibly as a result of compression during consolidation of the shale. The maximum size of pyritic lenses is about 12 mm.

Zoning of some disseminated pyrite grains is revealed by slight color variations. Rounded to subhedral cores within some pyrite grains may have resulted from recrystallization or growth zoning. Other grains exhibit euhedral cores within rims of pyrite which produce euhedral to rounded outlines.

Corroded grains of pyrite are seen in most polished sections, but the most advanced etching appears at and immediately above the Cu-Fe transition. Skeletal forms produced by severe corrosion are shown in Figure 10.

Chalcopyrite.—The only copper-bearing sulfide seen disseminated within the pyritic zone is chalcopyrite. Typically it occurs as irregular grains 15–30 μ in diameter. This mineral presumably accounts for most of the low copper content (less than 200 ppm) of the pyritic zone (Fig. 8).

Mineralogy of the Cupriferous Zone

Most of the copper of the cupriferous zone occurs as chalcocite with lesser amounts in the form of

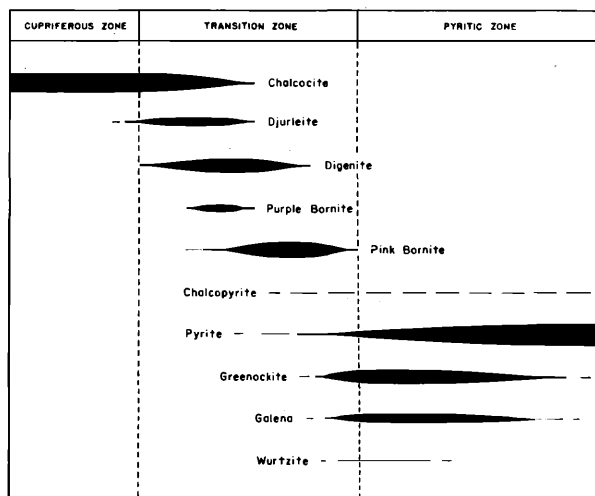
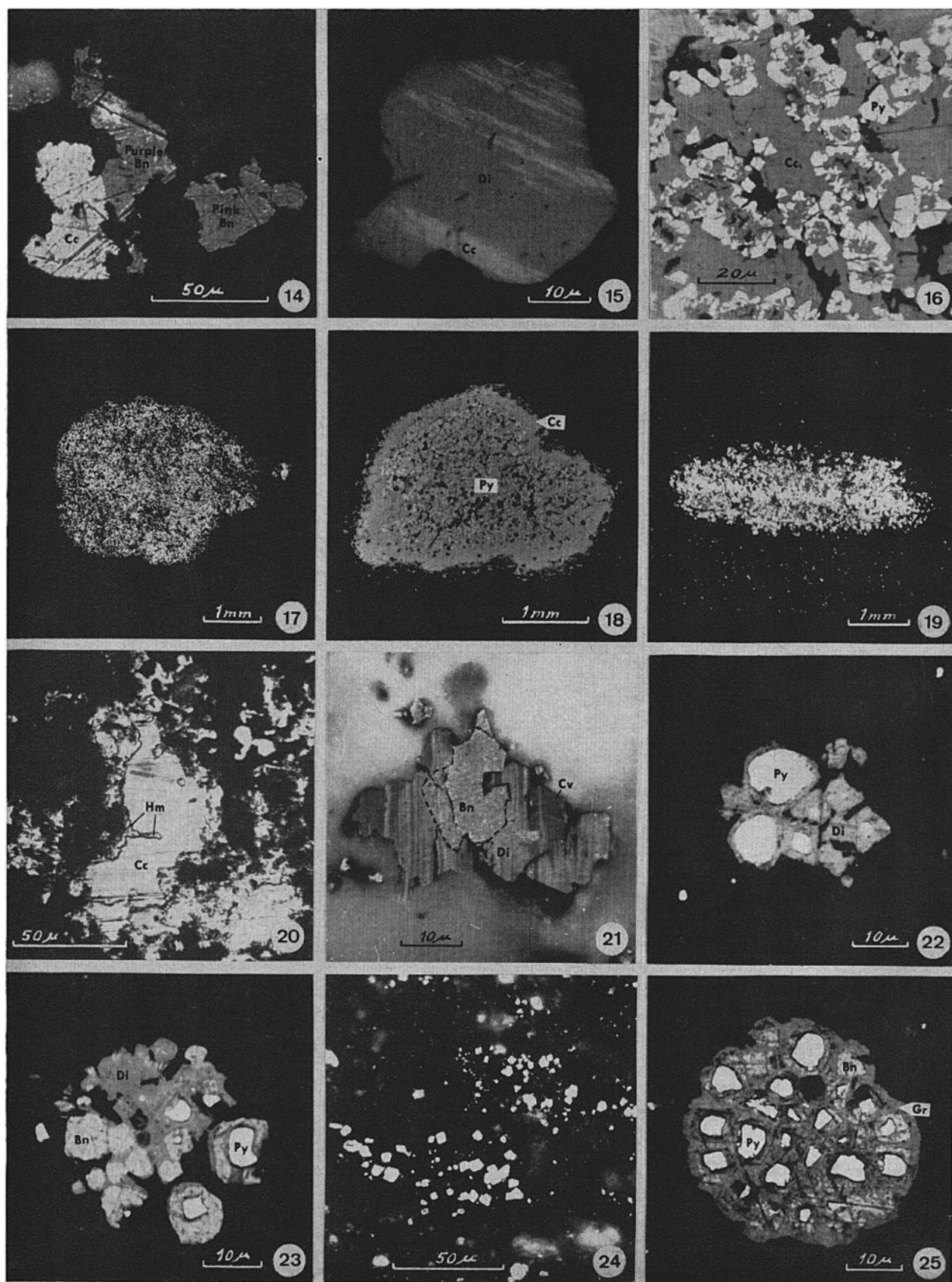


FIG. 13. Typical zonation of sulfides at the top of the cupriferous zone.



native copper. Very small amounts of copper-iron sulfides have been noted within the ore zone, particularly in and near fractures (Carpenter, 1963).

Chalcocite.—Chalcocite generally occurs as irregular grains disseminated through the rock, and tends to be concentrated along certain bedding planes. It is commonly intimately associated with chlorite, carbonate and organic matter, and probably replaces these shale constituents. The average size of chalcocite grains is 20–40 μ .

A minor amount of chalcocite is found as nodules along bedding planes (White and Wright, 1954, 1966). Small nodules (< 5 mm) are commonly spherical whereas larger nodules (5–20 mm) are normally lenticular in shape. The internal structure of the nodules typically consists of coarse interlocking grains. A few nodules seen inches below the Cu-Fe transition are composed of aggregates of very fine-grained pseudomorphs after pyrite euhedra. A rim of hematite is found around many but not all chalcocite nodules. It seems likely that these nodules were once pyritic and have been replaced by chalcocite. The hematite is presumably a by-product of the mineralization process.

A minor amount of chalcocite occurs as veinlets which both cross-cut and follow bedding planes. Most veinlets are only small fractions of an inch in width.

Native Copper.—Disseminations and interstitial fillings of native copper are rather common in lower portions of the ore zone. It is normally most abundant associated with chalcocite in dark grey beds. The highest stratigraphic occurrence of native copper in the lower Nonesuch is apparently related to the stratigraphic position of the Cu-Fe transition (Fig. 12). Thin sheets of native copper have been found along fractures and bedding planes of thinly laminated strata in some portions of the mine.

Hamilton (1967) has described native copper occurrences within a chloritic facies of the Lower Sandstone. This metallic copper was apparently deposited under reducing conditions existing in proximity to organic matter and chlorite. The scarcity of copper sulfides in the Lower Sandstone is presumably due to the lack of original iron sulfides.

Native Silver.—Native silver is a minor ore mineral, but is not uncommon in highly mineralized portions of the Parting Shale and in the Upper Shale of several drill core sections. Carpenter (1963) noted that it is very closely associated with native copper and commonly rims it. Native silver is also observed in cross-cutting fractures and along bedding planes, particularly in association with native copper.

Mineralogy of the Cu-Fe Transition

As seen in hand specimen, the top of the cupriferous zone is a narrow zone of disseminated sulfides in which chalcocite gives way upward to a subzone of bornite, then to very fine-grained pyrite. Conspicuous grains of yellow greenockite are commonly visible immediately above the bornite subzone.

Polished sections reveal a more complex array of copper-iron sulfides. The usual sequence observed is, in ascending order, chalcocite, digenite, bornite (two varieties), chalcopyrite (sparse), and pyrite. The typical overlap of these mineralogic subzones and the relative abundances of each sulfide are shown schematically in Figure 13. The complete transition from the cupriferous to the pyrite zone commonly occurs within a vertical distance of several inches, but varies from less than one inch where mineralization is strong to several feet where mineralization is weak. Sparse amounts of galena and wurtzite are locally associated with anomalous concentrations of greenockite at the top of the fringe. The number

FIG. 14. Association of chalcocite (Cc), purple bornite (in contact with chalcocite), and pink bornite (separated from chalcocite by gangue minerals). Reflected light, in oil.

FIG. 15. Interlaminated chalcocite (Cc) and digenite (Di). Reflected light, in oil.

FIG. 16. Pervasive replacement of gangue by chalcocite (Cc) which has also veined and replaced cores of pyrite euhedra (Py). Reflected light, in oil.

FIG. 17. Aggregate of pyrite euhedra located 2 feet above the Cu-Fe transition. See also Figs. 18 and 19. Reflected light, in air.

FIG. 18. Partial replacement of an aggregate of pyrite euhedra (Py) by chalcocite (Cc) at the level of the Cu-Fe transition. Similar structures at the same stratigraphic horizon in neighboring drill core sections were found to be pyritic (Fig. 17) or chalcocitic (Fig. 19). Reflected light, in air.

FIG. 19. Complete replacement of an aggregate of pyritic euhedra by chalcocite located 2 feet below the fringe surface. See also Figs. 17 and 18. Reflected light, in air.

FIG. 20. Hematite (Hm) rimming and embaying a chalcocite grain (Cc). Reflected light, in oil.

FIG. 21. Interlamination of digenite (Di) and covellite (Cv) in association with pink bornite (Bn). Reflected light, in oil.

FIG. 22. Partial replacement of pyrite euhedra by digenite (Di), leaving remnant cores of pyrite. Reflected light, in air.

FIG. 23. Successive replacement of pyrite by bornite (Bn), then digenite (Di), leaving remnant cores of pyrite grains (Py). Reflected light, in oil.

FIG. 24. Pseudomorphs of bornite after pyrite euhedra. Reflected light, in oil.

FIG. 25. Partial replacement of a framboidal group of pyrite grains by bornite (Bn) has left remnant cores of pyrite (Py). Greenockite (Gr) rims the entire structure and projects into it along grain boundaries. Reflected light, in oil.

of sulfides together with their intricate textures make the mineralogy of the Cu-Fe transition rather complex in comparison with the mineralogies of the barren pyritic shales above and the mineralized shales below.

Chalcocite.—Most of the chalcocite at the fringe occurs as separate isolated grains whose abundance decreases sharply as iron-rich phases increase in abundance in the upper portions of the fringe. Chalcocite most commonly has mutual boundaries with digenite and bornite (Fig. 14). Bornite and digenite occur as inclusions in chalcocite, but chalcocite is rarely included in other copper-bearing sulfides. Chalcocite is also found as lamellar oriented intergrowths with digenite (Fig. 15) and less commonly with covellite.

Pseudomorphs of chalcocite after pyrite euhedra are common in the Cu-Fe transition, although pseudomorphs of iron-bearing copper sulfides after pyrite are generally more prevalent. In many places chalcocite has replaced the outer rims of pyrite grains, leaving unreplaced cores of pyrite. In other cases chalcocite has preferentially replaced the cores of pyrite grains (Fig. 16). An intermediate zone of digenite and/or bornite may exist between the chalcocite and pyrite phases. Pseudomorphs of chalcocite after pyrite euhedra are not observed within the cupriferous zone proper according to Wiese (1960).

Aggregates of pyrite euhedra may also show textures resulting from successive stages of replacement by chalcocite (Figs. 17, 18 and 19). Halos of hematite surround some aggregates replaced by chalcocite. Hematite also is found as dusty disseminations adjacent to chalcocite and as rims enclosing disseminated grains of chalcocite (Fig. 20).

Djurleite.—Djurleite ($\text{Cu}_{1.96}\text{S}$) may be a common sulfide in the Cu-Fe transition if not within the cupriferous zone proper. Optical identification of djurleite is difficult because of its close similarity to chalcocite. Identification by X-ray powder diffraction methods is also hindered by the very fine grain size of sulfides and the similarity of chalcocite and djurleite diffraction patterns (Roseboom, 1962, 1966). Readable diffraction patterns were obtained with some difficulty from chalcocite-like material in all well-mineralized beds of the cupriferous zone. The most diagnostic djurleite reflection ($d = 3.386 \text{ \AA}$) was detected only in samples from the base of the Cu-Fe transition.

Digenite.—The presence of digenite, formerly identified as "blue chalcocite" at White Pine (Wiese, 1960; Carpenter, 1963), is suggested by its close association with chalcocite and bornite. Confirmation of the digenite structure has been made by X-ray diffraction techniques.

Grains of digenite are similar in size and shape to disseminated chalcocite grains and are either isolated or occur in contact with other sulfides. Digenite occurs most commonly in mutual contact with bornite. Less common associations are digenite-covellite, digenite-chalcocite, digenite-chalcopyrite, and digenite-pyrite. Covellite normally forms lamellar intergrowths with digenite (Fig. 21).

Digenite pseudomorphs after pyrite commonly contain and contact remnant cores of pyrite (Fig. 22), but normally bornite forms an intermediate zone between the pyrite core and digenite rim (Fig. 23).

Bornite.—Two distinct varieties of bornite occur within the White Pine fringe. On freshly buffed polished surfaces they are both pink in color, but after a few hours exposure to air one becomes decidedly purplish while the other retains its pinkish tinge for an extended period of time. The rapidly tarnishing phase, here called "purple bornite," is almost invariably adjacent to chalcocite. The "pink bornite" is never observed in contact with chalcocite even though the two bornites are commonly found in the same local area of a polished section (Fig. 14). Where disseminations of fine-grained bornite (e.g., pseudomorphs after pyrite) are found near relatively coarse-grained chalcocite, those bornite grains nearest chalcocite are normally purple whereas those more distant are pink. On the other hand one may find completely random mixtures of purple and pink varieties of bornite within a field of bornite pseudomorphs after pyrite.

The most common texture of purple bornite is a mutual boundary contact with irregular grains of chalcocite. Purple bornite pseudomorphs after pyrite are also very common. Purple bornite-digenite contacts are rare, but purple bornite commonly forms lamellar intergrowths with covellite in some polished sections studied.

Pink bornite commonly occurs as irregular grains of the same size as chalcocite and digenite grains. It is also very common as pseudomorphs after pyrite euhedra (Figs. 23, 24 and 25) and as rims around skeletal pyrite grains (Fig. 26). Digenite, its most common associate, generally forms mutual contacts with this bornite. In some polished sections digenite is visible as apparent inclusions in bornite grains; the opposite relationship has not been recorded. Very fine veinlets filled with digenite join the apparent inclusions to thin rims of digenite surrounding the bornite grains. These rims of digenite are uniform in thickness even in cases in which the bornite grains are smaller in volume than the enclosing rims. It is concluded that the digenite rims could not have exsolved from the bornite and that digenite is paragenetically younger than the bornite.

Consistent association of purple bornite with chal-

cocite suggests that it is a copper-rich variety. The pink phase may be more sulfur and/or iron-rich as suggested by its close association with sulfides containing more sulfur and iron than bornite. Artificial annealing of samples containing pink bornite caused exsolution of chalcopyrite (Fig. 27) similar to that described for low temperature bornites (Brett and Yund, 1964). The average volume of exsolved chalcopyrite is estimated to be 5–10 percent of the bornite volume. This strongly suggests that the pink bornite is similar to “sulfur-rich” x-bornites described in recent literature (Yund and Kullerud, 1966; Skinner, 1966). Although temperatures reached 150° C during the present annealing experiments, Brett and Yund found that chalcopyrite exsolves from these anomalous bornites on annealing from 75° C at cooling rates of less than 3° C per day. The low temperature of pink bornite formation suggested by these data is discussed in a later section of the paper.

Chalcopyrite.—In most fringe profiles, chalcopyrite is remarkably sparse, rarely occurring in amounts above the traces that can be found at higher stratigraphic levels in the pyrite zone. Where present, it commonly occurs as irregular disseminated grains 20–40 μ in size. The most common contact associations are chalcopyrite-bornite and chalcopyrite-digenite. Pseudomorphs of chalcopyrite after pyrite euhedra can be found in association with digenite, bornite, covellite, and remnants of pyrite. The lowest observed occurrence of disseminated chalcopyrite was at the pink bornite subzone.

Pyrite.—The Cu-Fe transition is completed with the appearance of pyrite above the bornite subzone. The first pyrite encountered above the cupriferous zone occurs as minute specks disseminated through the fringe as low as the digenite subzone or as remnant cores of pyrite euhedra rimmed by copper-bearing sulfides at or about the same mineralogic subzone. Isolated specks of pyrite presumably represent etched remnants of former pyrite euhedra.

At the top of the bornite subzone, pyrite increases rapidly in abundance, and corrosion of euhedra is noticeably diminished. Uncorroded euhedra may be found at the top of the fringe, but in many cases etching is seen for several inches into the pyritic zone above. Pyrite occurrences at higher levels have been described earlier.

Other Sulfides

Greenockite.—Greenockite (CdS, hexagonal) is usually found as disseminations in anomalous amounts (1) in a narrow subzone several inches or feet in thickness immediately above the Cu-Fe transition, and (2) in the lower portions of the No. 61 bed where that bed is pyritic. Where the No. 61

horizon is mineralized, it contains no greenockite and concentrations of CdS are found only at the fringe in strata overlying the No. 61 bed.

In the fringe zone, greenockite is commonly associated with pyrite and chalcopyrite (Fig. 28), less commonly with bornite (Fig. 25) and digenite, and seldom with chalcocite. It occurs chiefly as irregular grains which in some cases enclose pyrite euhedra. It also rims other sulfides, and rarely occurs as pseudomorphs after pyrite euhedra (Fig. 29). Greenockite grains are usually 20–40 μ in size, but commonly attain dimensions of 200 μ or more where the mineral replaces carbonate gangue. Calcite nodules such as occur in the No. 44 bed are particularly favorable hosts for disseminated greenockite.

Wurtzite.—Wurtzite (ZnS, hexagonal) has been identified through X-ray powder diffraction studies of material from the top of the No. 61 bed. Microchemical and X-ray fluorescence analyses gave positive results for zinc and negative results for cadmium.

In hand specimen wurtzite closely resembles greenockite in habit and occurrence, but it has a distinct creamy-white color in contrast to the yellow of greenockite. Sparsely disseminated grains of a mineral having the properties of wurtzite have also been seen in polished sections of the fringe zone.

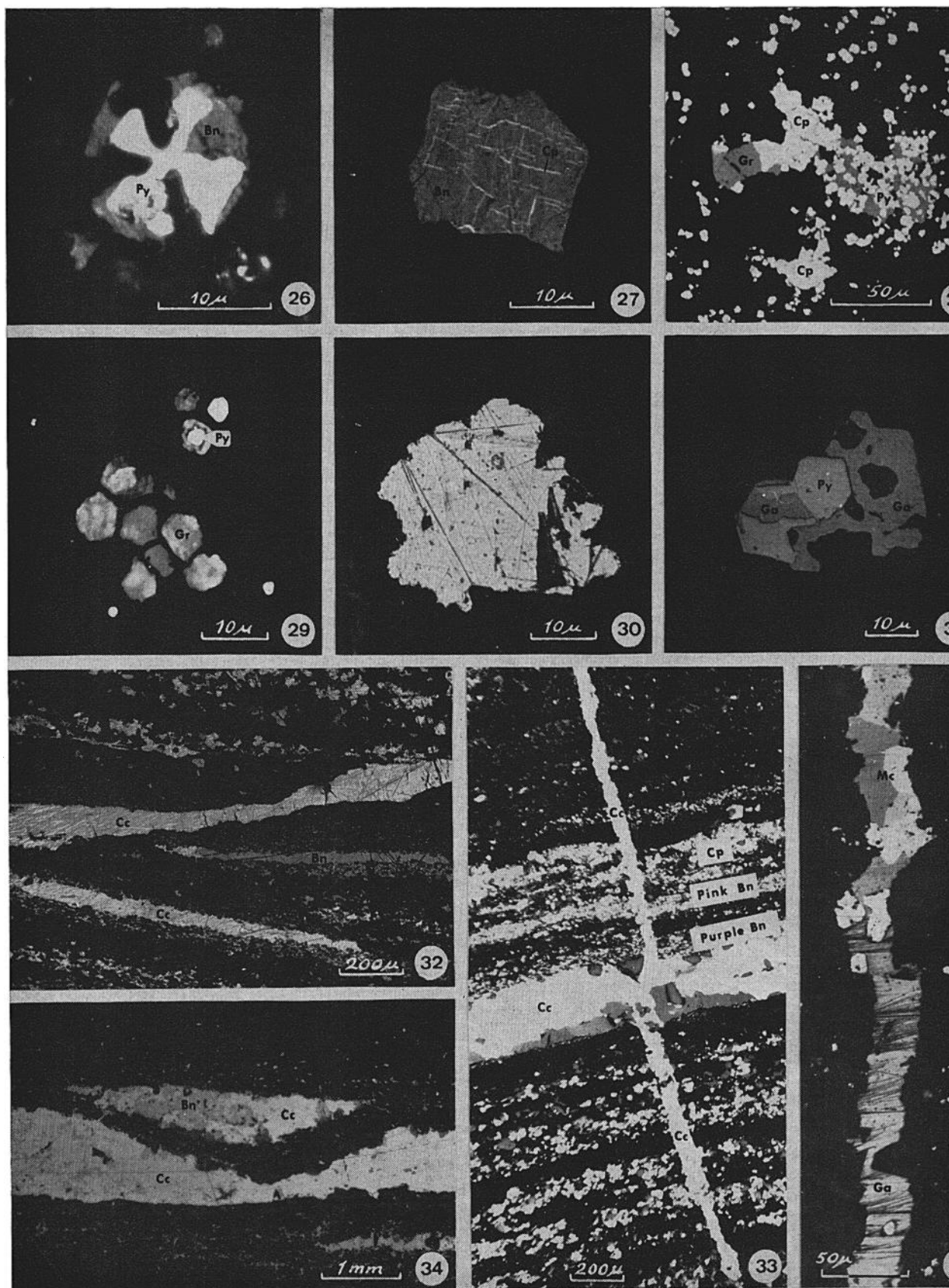
Galena.—Anomalous concentrations of galena as sparsely disseminated grains 10–50 μ in size (Fig. 30) occur at the top of the fringe and also in the pyritic shale in and above the No. 61 bed. Thus it closely parallels greenockite in occurrence. Galena is found within greenockite grains and as partial replacements of pyrite (Fig. 31). Pseudomorphs of galena after pyrite euhedra are rare.

Covellite.—Several polished sections of fringe samples show covellite, particularly in association with chalcocite (djurleite?), digenite and bornite (Fig. 21). It is absent in many fringe profiles and its distribution within the cupriferous zone is not well known. It does rim greenockite, one of the latest fringe sulfides. This late age relationship and the high sulfur composition of covellite indicate a possible association with the sulfur-rich late vein mineralization described by Carpenter (1963) or a late supergene alteration of copper-bearing sulfides.

Molybdenite.—Molybdenite has been found on a slickensided fracture surface in one area of the mine workings. A few isolated grains also occur in calcite-filled veinlets adjacent to the main fracture. Samples examined by the writer show disseminated pyrite and chalcopyrite within bedding, suggesting that the samples were obtained from the pyritic zone.

Vein Mineralization

Four distinct types of vein mineralization are recognized in the lower Nonesuch strata: (1) bed-



ding seams, (2) hairline veinlets, (3) early veins, and (4) late veins. Each type forms only a very minor portion of the total mineralization at White Pine. The so-called late veins have been described previously by Carpenter (1963).

Bedding seams, hairline veinlets, and early veins contain sulfides which closely resemble those of the fringe in both vertical zoning and paragenesis. That is, one observes the ascending array chalcocite $\rightarrow$ digenite $\rightarrow$ bornite $\rightarrow$ (chalcopyrite) $\rightarrow$ (pyrite) in which copper-rich sulfides are paragenetically younger than iron-rich sulfides. Chalcopyrite and pyrite are rare. Subzones of copper-bearing sulfides in these three types of veins typically occur several inches to feet stratigraphically above the corresponding subzones of disseminated sulfides within the adjacent bedding.

Bedding seams are usually a few millimeters in thickness and commonly extend many tens to hundreds of feet along bedding planes. They may cross-cut bedding at very shallow angles and terminate abruptly (Fig. 32). Quartz and calcite fillings preceded sulfide mineralization in many cases. The most common sulfides are coarse-grained chalcocite within the cupriferous zone and chalcocite-bornite at the level of the fringe. Chalcocite-filled seams normally cut the bornite subzone of disseminated sulfides within bedding, and bornite-filled seams may intersect several inches of the lowermost pyritic zone. In these cases, the disseminated sulfides within the beds above and below the seam are less and less copper-rich outward into the walls (Fig. 33). Soft sediment deformations of the bedding seam walls (Fig. 34) suggest an early diagenetic age for the development of these fractures.

Figure 33 also shows a hairline veinlet extending outward from a bedding seam. These fractures are normally less than 1 mm in width and cross-cut only a few inches of shale. They may be associated with bedding seams, as illustrated, or be completely independent of them. At the level of the fringe, hairline fractures commonly contain chalcocite and bornite

for distances of several inches into the base of the pyritic zone.

Larger fractures which also contain sulfides related in mineralogy and paragenesis to the fringe are called early veins. This term is employed to distinguish them from late veins which are similar in magnitude but exhibit a paragenetic trend from copper-rich to iron-rich phases. In contrast to hairline veinlets, early veins typically cross-cut thick sections of shale. They commonly occur as systems of veins along fracture zones which may be mapped for tens to hundreds of feet laterally within the mine. The width of early veins is normally less than 10 mm.

Copper mineralization in early veins is usually carried several feet into the pyritic zone along these more extensive fractures and also extends laterally from the vein into the adjacent wall rock. In a fracture zone where the mineralization of bedding may be continuous between fractures, the fringe of disseminated sulfides is displaced upward relative to the fringe position in nearby unfractured rock (e.g., drill cores 27 $\frac{3}{8}$ D $\frac{1}{2}$ and 27 $\frac{3}{8}$ D $\frac{3}{4}$, Fig. 3).

Greenockite, wurtzite, and galena are commonly found in small amounts in cross-cutting hairline veinlets and early veins, and rarely in bedding seams. These vein sulfides occur immediately above the zone of copper-bearing vein sulfides in a vertical array parallel to the distribution of Cd, Pb, and Zn sulfides within bedding. Marcasite has been found with galena in hairline fractures within the pyritic zone (Fig. 35).

The most reasonable explanation for vein sulfides involves a close genetic relationship with the disseminated mineralization within bedding. If cross-cutting and bedding plane fractures developed toward the end of the main mineralization period, the final very weak ore solutions would tend to migrate upward and along these channelways rather than through the shale itself. Small amounts of copper which penetrated the shale in this manner would have been precipitated by reaction with sulfide in the pyritic zone within short distances above the

FIG. 26. Corroded grains of pyrite (Py) showing partial replacement by bornite (Bn). Reflected light, in oil.

FIG. 27. Blades of chalcopyrite (Cp) exsolved from bornite (Bn) by annealing. Reflected light, in oil.

FIG. 28. Association of greenockite (Gr) and chalcopyrite (Cp) with pyrite (Py) in unmineralized shale above the Cu-Fe transition. Reflected light, in oil.

FIG. 29. Greenockite pseudomorphs (Gr) after pyrite euhedra (Py). One grain contains a core of pyrite. Reflected light, in oil.

FIG. 30. Irregular grain of galena showing triangular cleavage pits. Reflected light, in oil.

FIG. 31. Pyrite grains (Py) showing replacement by galena (Ga). Reflected light, in oil.

FIG. 32. Chalcocite-bornite bedding seams transecting bedding at moderately low angles. The mineralization of the central seams passes from purple bornite (Bn) to chalcocite (Cc) in the direction approaching the two chalcocite seams. Reflected light, in air.

FIG. 33. Conformable bedding seam intersected by a cross-cutting veinlet. Vein mineralization consists of chalcocite (Cc) and unreplaced quartz and carbonate. Mineralization within bedding is zoned about the bedding seam with copper-rich sulfides next to the chalcocite of the bedding seam. No zoning is apparent with respect to the cross-cutting veinlet. Reflected light, in air.

FIG. 34. Irregular wall of bedding seam indicative of soft-sediment deformation. Reflected light, in air.

FIG. 35. Cross-cutting veinlet filled with galena (Ga) and marcasite (Mc, various tones of gray due to anisotropy). Reflected light, polars crossed to 80°, in oil.

fringe position in adjacent unfractured ground. The resultant vein mineralization would be minor in proportion to the mineralization within bedding and would be paragenetically similar to sulfides in the fringe. These are typical characteristics of vein sulfides described above.

It is unlikely that fractures of these types existed during the main phase of mineralization. Although analysis of fractures remains an active study at White Pine, evidence to date does not indicate any obvious relationship between the density or size of fracture systems and the *regional* configuration of the fringe surface. Hence it seems that copper distribution through the shale was not controlled by fractures. Moreover it was shown earlier in this paper that a strong correlation exists between the fringe height in the Nonesuch strata and the amount of copper in the No. 21 + 23 beds. That relationship would be difficult to explain if copper has been guided through the shale along fracture systems known at this time.

Elemental and Mineralogic Distributions Across the Cu-Fe Transition

Earlier in this paper evidence was given that the mineralization at White Pine, Michigan, resulted from additions of copper and minor amounts of silver to basal units of the Nonesuch Shale. Small concentrations of cadmium, lead (and zinc), appear to have accumulated above the ascending mineralization front. Sulfur and iron from original pyrite were intimately involved in the mineralization process, but it is not immediately obvious whether these elements experienced any appreciable redistribution, additions, or losses, in particular across the Cu-Fe transition. An analytical study was designed to determine more precisely the distributions of these seven key elements.

All samples analyzed consisted of continuous shavings from complete vertical drill-core sections. Individual samples represented stratigraphic thicknesses varying from 1.5 inches to 2 feet. Three types of sample arrays were selected for investigation:

- 1) Two continuous profiles which transect the entire cupriferous zone and extend into the pyritic zone as far as the No. 65 bed (141 samples),
- 2) Profiles representing stratigraphic thicknesses of 3–10 feet in five different drill core sections transecting the Cu-Fe transition (83 samples), and
- 3) Samples of the complete No. 46 bed of 12 different drill cores: 5 cupriferous, 3 pyritic, and 2 which contain the Cu-Fe transition within the No. 46 bed (12 duplicated samples).

Elemental distributions from the vertical sampling arrays (1) and (2) are adequate for gross descriptions of concentrations in the lower Nonesuch Shale, but distributions resulting from the mineralization process alone are not entirely obvious because of normal variations in total sulfide content *between* stratigraphic levels. This problem is minimized in the sample arrays confined to selected stratigraphic horizons such as group (3) because the sulfide content of any particular bed tends to be rather uniform over broad regions. Consequently minor differences in elemental concentrations within a given bed cut by the Cu-Fe transition can be more clearly attributed to the fringe mineralization and hence are likely to be more meaningful than distributions in vertical arrays such as groups (1) and (2).

Elemental analyses were made by X-ray fluorescence techniques (Birks, 1959; Liebhafsky et al., 1960; Adler, 1966) for Cu, Ag, S, Cd, Pb, Zn, and Mo, using pelletized samples of pulverized rock specimens. Total iron was determined by colorimetry using the method of Shapiro and Brannock (1962).

In addition to the elemental investigations, some comparative studies were made of chlorite, carbonate, and organic matter contents across the fringe. These constituents are commonly associated with chalcocite and native copper within the ore zones and may have been partially affected by mineralization. Relative chlorite contents were estimated from the sum of (001), (002), (003) and (004) X-ray diffraction peaks according to the method of Wiese (1960). The samples used in this study consisted of the 12 duplicated samples of group (3) above which had been previously pelletized for X-ray fluorescence analysis. The iron contents of the chlorite were calculated from the ratios of the (002) and (001) peak intensities (Wiese, 1960).

Carbonate was determined by the evolved volume of CO₂ on treatment of powdered samples with dilute HCl. Total organic matter was found by weight loss after combustion, with corrections for carbonate content. A soluble organic content was determined by extraction with a benzene-methanol mixture (Hoering, 1965). The samples used for carbonate and organic analyses consisted of eight selected specimens from specific stratigraphic horizons cut by the fringe surface (Table 4).

Results From Elemental Analyses

The following description is a condensation of the results and interpretations from the elemental analyses. A more complete account is given by Brown (1968). Figures 36 and 37 have been chosen as representative of analytical profiles derived from this study.

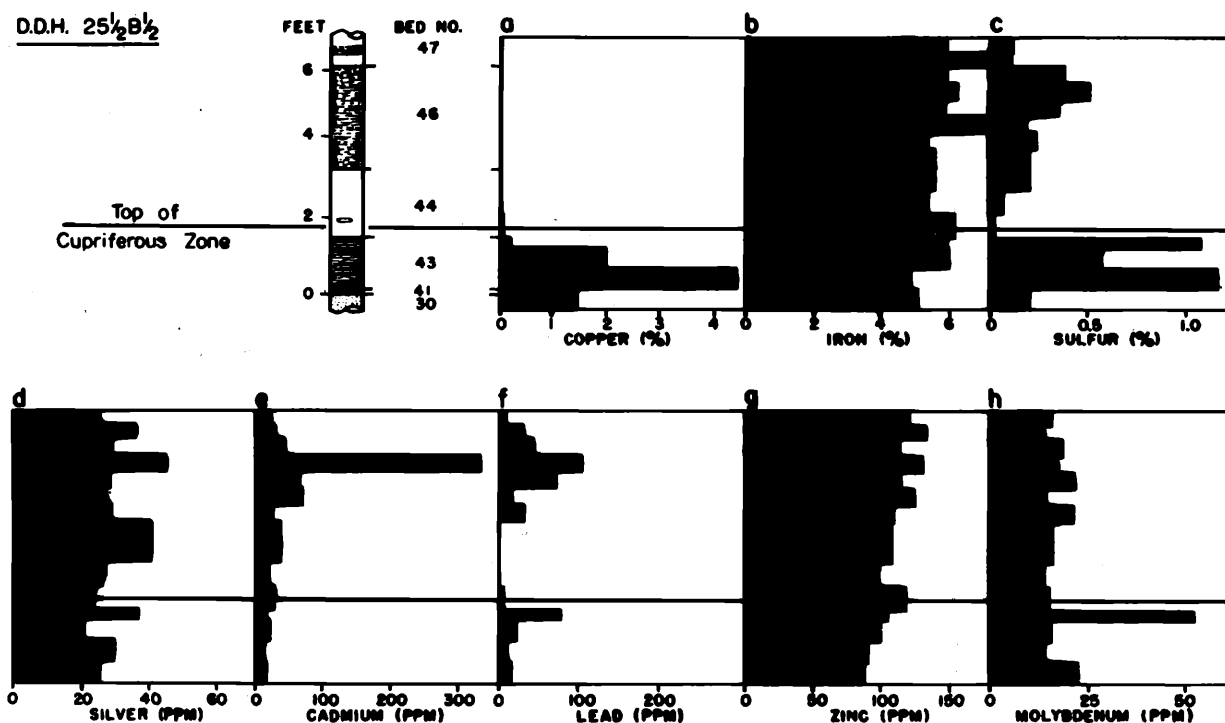


FIG. 36. Elemental distributions in the Upper Shale section of drill core 25½B1.

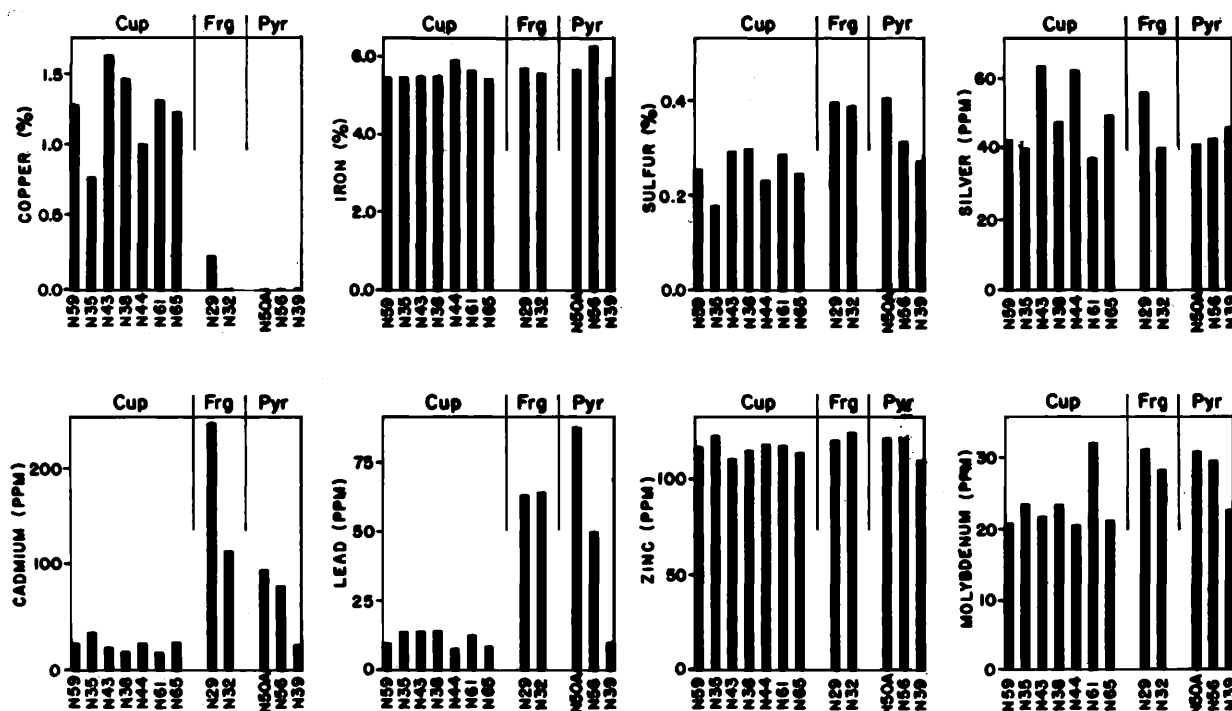


FIG. 37. Elemental concentrations in 12 duplicated samples of the No. 46 bed. Cup = cupriferous samples, Frg = samples from the Cu-Fe transition, and Pyr = pyritic samples.

Copper-Silver.—Copper and silver distributions in the lower Nonesuch Shale have been discussed in several publications (see earlier list of related investigations) and require little further comment in

the present paper. Copper values shown in the short vertical profile of Figure 36 and the lateral sample array of Figure 37 demonstrate the typically abrupt cut-off in values at the fringe. Concentrations

within the cupriferous zone are almost invariably greater than 200 ppm even in poorly mineralized beds. Copper concentrations within the pyritic zone rarely exceed 200 ppm. A few anomalously high copper values obtained from samples located short distances above the Cu-Fe transition in some sample arrays can be attributed to mineralization which had been previously identified in veins.

Silver shows a rather uniform distribution (weighted average of 30 ppm) throughout the basal Nonesuch with a few sporadically high values (up to 85 ppm) in the lower portions of the cupriferous zone where native silver occurs in association with native copper.

Iron.—No significant difference was found between the iron contents of the pyritic and cupriferous shales (Fig. 37). This result may seem unusual because pyrite is found above but not below the fringe. This surprising result is explained in part by the fact that only a minor portion of the total iron of the shale occurs in pyrite. The remainder exists as an essentially inert constituent of silicates and oxides which were apparently unaffected by the mineralization front. Secondly, the iron from original pyrite replaced at the mineralization front appears to have been re-deposited locally as dusty disseminations of hematite and/or by incorporation into diagenetic chlorites.

Sulfur.—Two previous studies of sulfur distributions across the fringe (Brown, 1965; White and Wright, 1966) have suggested that the cupriferous zone is sulfur deficient relative to the unmineralized shale. This conclusion is further supported by results obtained in this study. Figure 37 shows an average of 0.29% S (range 0.20–0.33%) among seven cupriferous samples, an average of 0.37% (range 0.30–0.46%) among three pyritic samples, and 0.44–0.45% in the two fringe samples. The apparent lower content of sulfur observed between the No. 43 and No. 46 beds of Figure 36 is probably a result of normal between-bed variations and is not necessarily related to the fringe position in any obvious manner.

A deficiency of sulfur in beds beneath the Cu-Fe transition can be explained by both syngenetic or epigenetic processes. The activity of sulfate-reducing bacteria indirectly responsible for the precipitation of syngenetic sulfides may have been inhibited by the presence of copper in the bottom muds (although Temple and LeRoux (1964) have demonstrated that the toxicity of copper may be exaggerated). It is equally possible that sulfur was lost at the mineralization front during epigenetic replacement of original pyrite by copper-bearing sulfides. In fact, textural evidence (Fig. 10) indicates that pyrite was etched at and immediately above the

fringe. It would be remarkable indeed if some portion of the sulfur released from pyrite did not escape re-deposition at the mineralization front, especially if there were any appreciable upward flow of interstitial solutions.

Two alternative explanations for the indicated sulfur deficiency beneath the fringe are worthy of consideration. The activity of sulfate-reducing bacteria may have continued during and perhaps even after mineralization, adding further sulfur to the shale. Such late bacterially-controlled contributions may have been greater within the pyritic shale where possible copper toxicity would have been negligible in comparison with the mineralized shale. The other suggestion involves a possible relationship between sulfur distributions and late vein mineralization which is characterized by additions of sulfur (Carpenter, 1963). However, these fracture-controlled additions, if any, were probably of no consequence as the total amount of sulfides in late veins is negligible compared to the amount of disseminated sulfides within bedding, and any contributions of sulfur to the adjacent wall rock would probably have been proportionately small.

Cadmium-Lead.—The majority of the lower Nonesuch samples are characterized by cadmium values of less than 40 ppm and lead values of less than 30 ppm. Anomalous concentrations of these two metals are typically found (1) in a narrow zone immediately overlying the Cu-Fe transition, and (2) in pyritic portions of the No. 61 bed. Cadmium commonly attains values of 200–300 ppm at the fringe (maximum of 1,110 ppm in this study). Lead concentrations at the level of the fringe are typically 75–200 ppm (maximum value of 675 ppm). Such concentrations are clearly exhibited in Figures 36 and 37. The low values in the middle of the cadmium and lead profiles of Figure 36 occur in the No. 44 bed which is normally a sulfide-poor horizon.

These anomalous concentrations contrast with weighted averages of 28 ppm and 10 ppm Pb for samples of the cupriferous zone, and weighted averages of 35 ppm Cd and 11 ppm Pb for most samples above the fringe. The distribution of lead differs significantly from cadmium above the No. 61 bed where high lead concentrations (weighted average of 94 ppm) continue into the No. 63 and No. 65 beds. The cadmium content of the pyritic beds above the No. 61 horizon (weighted average of 27 ppm) compares closely with that of other pyritic beds.

Zinc-Molybdenum.—Unlike cadmium and lead, zinc and molybdenum show no distinct anomalies at the level of the fringe. Zinc in the few grains of wurtzite normally seen in the Cu-Fe transition ap-

parently does not exist in sufficient amounts to cause an obvious anomaly at the top of the cupriferous zone. Zinc values from all samples analysed including both mineralized and unmineralized shale are closely grouped around a median value of 106 ppm. Slightly anomalous zinc concentrations (up to 172 ppm) were found in the No. 61 and No. 65 beds. This agrees with the common occurrence of sparsely disseminated zinc sulfide in those strata.

Molybdenum exhibits no definite contrasts across the lower Nonesuch shale. Most samples contain 20–30 ppm Mo. The maximum value obtained was 56 ppm from a sample of the No. 26 bed. The lack of molybdenum concentrations in any portion of the basal Nonesuch is supported by mineralogic investigations which have indicated that the shale is devoid of molybdenum-bearing minerals.

Results from Mineralogic and Organic Analyses

Chlorite.—The relative chlorite contents listed in Table 4 are not detectably different among the seven cupriferous samples (194.0–277.5; average 227.5), the two fringe samples (201.5–240.5; average 225.0), and the three pyritic samples (187.5–250.0; average 226.0). From these limited data, it is tentatively concluded that there is no significant difference in the chlorite contents of the mineralized and unmineralized shale. This conclusion conflicts with the observation that chlorite has been replaced by copper sulfides (Weise, 1960) unless an originally greater chlorite content in the present cupriferous zone was reduced during mineralization to values which by chance resemble those of the unmineralized shale.

The relative iron contents of chlorite (Table 4) also show no marked contrast across the Cu-Fe transition: cupriferous samples (2.41–2.80; average 2.59), fringe samples (2.38–2.51; average 2.45), and pyritic samples (2.46–2.73; average 2.57).

Carbonate.—Table 5 indicates that in all four samples from the No. 47 bed the carbonate content is low and remarkably uniform in value (0.081–0.098%). In the second array of four samples taken from the contact between the No. 46 and No. 47 beds, the two cupriferous samples are noticeably deficient in carbonate (0.075–0.16%) in comparison to the two pyritic samples (0.43–0.43%). The low carbonate contents of the mineralized samples probably results from partial replacement of calcite by copper-bearing sulfides.

Organic Matter.—With one exception, the analyses for total organic matter (0.038–0.39%) and soluble organic matter (0.039–0.079%) show no clear contrast between the pyritic and cupriferous samples (Table 5). Sample 7 from the No. 47 bed contains about three times as much total organic

Table 4. Estimates of relative chlorite contents (sum of intensities of (001), (002), (003) and (004) diffraction maxima) and relative iron contents of chlorite (ratio of (002)/(001) diffraction maxima) for 12 duplicate samples of No. 46 bed. Cup = cupriferous, Pyr = pyritic, and Frg = fringe samples

Sample	Relative Chlorite Contents	Relative Iron Contents in Chlorite
N35	263.0	2.41
N38	196.5	2.45
N43	277.5	2.50
N44	198.5	2.58
N59	226.0	2.69
N61	238.5	2.68
N65	194.0	2.80
N29	201.5	2.51
N32	248.5	2.38
N39	187.5	2.51
N50A	250.0	2.46
N56	240.5	2.73

matter as the other three samples of the No. 47 bed. No particular explanation is offered for this single anomaly.

Models of the Mineralization Process

Introduction

The configuration and zoning of the White Pine copper deposit suggest that the lower Nonesuch mineralization resulted from an influx of copper from the underlying Copper Harbor Conglomerate. Copper deposition apparently took place by reaction with indigenous pyrite along an upward advancing replacement front and by reduction to native copper in the presence of organic matter. The following is a discussion of plausible models of this mineralization process.

Two fundamental mechanisms are considered: (1) infiltration (the transport of copper in solution by bodily flow of the ore solution from the conglomerate into the overlying shale), and (2) diffusion (the migration of dissolved copper through a stationary pore solution due to a chemical potential gradient).

Ultimate Source of Copper

A reasonable and adequately large source of copper must have existed beneath the Nonesuch in order to suggest that either diffusion or infiltration processes could account for the White Pine mineral-

Table 5. Carbonate, total organic and soluble organic contents of samples distributed laterally across the Cu-Fe transition

Stratigraphic Unit Sampled	Sample No.	Zone	Carbonate as Calcite (%)	Total Organic Matter as C (%)	Soluble Organic Matter (%)
No. 47 Bed	7	Cuprif-ferous	0.098	0.311	0.059
	8		0.081	0.105	0.045
	5	Pyritic	0.087	0.092	0.058
	6		0.098	0.088	0.039
Contact of No. 46-47 Beds	1	Cuprif-ferous	0.163	0.386	0.070
	4		0.075	0.310	0.070
	2	Pyritic	0.431	0.355	0.079
	3		0.428	0.372	0.074

ization. To fulfill this condition, the following sources are suggested: (1) latent cupriferous solutions generated during the volcanic activity which ceased only a short time before Nonesuch deposition, (2) conventional magmatic hydrothermal solutions, and (3) extraction of copper from underlying extrusives and/or sediments by reaction with interstitial waters, a process akin to that suggested by Helgeson (1967).

The transport of copper from these potential sources to the base of the Nonesuch probably occurred by fluid flow of pore solutions within the highly permeable Copper Harbor formation. White and Wright (1966) suggested that the solutions were copper-bearing connate waters which were forced toward surface through aquifers such as the Copper Harbor conglomerate during compaction of the Keweenaw sedimentary-volcanic pile. D. E. White (1968) favors re-cycling of a copper-bearing brine by heat-induced convection.

Timing of Mineralization

A syngenetic origin for the White Pine mineralization is excluded on the basis of previously published evidence (White and Wright, 1966; Hamilton, 1967) and further diagnostic evidence presented earlier in this paper.

The major phase of mineralization apparently took place after the formation of early diagenetic pyrite and chlorite, which are replaced by copper-bearing sulfides. A minimum relative age can be established from the observation that the upper surface of the cupriferous zone is offset by the White Pine fault, showing that mineralization preceded that structural event which occurred in a lithologically competent shale. Further definition of the mineralization

per transport through the shale would be most period is difficult. One can only suggest that cop-favorable during the early history of the Nonesuch sediment when porosity and permeability were great-est. White and Wright (1966) have suggested that the notable paucity of pseudomorphs after pyrite within the cupriferous zone may indicate that cop-per sulfides have replaced an early primitive iron sulfide in the lower shale horizons.

A very small portion of late-stage copper was apparently guided upward along minor fractures producing local elevations in the fringe surface. However the lack of any obvious correlation of the regional configuration of the fringe surface with the size or density of fractures suggests that mineraliza-tion was essentially complete before the shale was fractured.

Chaudhuri and Faure (1967) attempted to date the White Pine mineralization by lead isotope meth-ods. Their data suggest an age of less than 720 m.y. for the disseminated chalcocite compared to an age of $1,075 \pm 50$ m.y. for the host shale. While this estimate does support White and Wright's postulate that the mineralization is epigenetic, Chaudhuri and Faure stress that their interpretation of the lead isotope data is very tentative.

Pressure-Temperature Environment

The maximum pressures at the time of mineral-ization were probably due to the lithostatic load of overlying sediments. Ensign et al. (1968) esti-mate the thickness of the Nonesuch and Freda for-mations combined was 8,000 + feet. This figure suggests a maximum lithostatic pressure of about 650 bars and a minimum hydrostatic pressure of about 240 bars. Such low static loads would have

negligible effects on the mineralization processes (Helgeson, 1968). On the other hand, directed pressures such as those due to the expulsion of connate waters during compaction may have contributed significantly to the transport of copper.

Temperatures within the Nonesuch Shale were probably never severe at any time. This is evidenced by the lack of metamorphic effects and wall-rock alteration. If the Nonesuch were buried by the Freda Sandstone during mineralization, a temperature of 120° C above surface conditions might have occurred assuming a geothermal gradient of 1° C/100 feet of burial. Higher temperatures may have resulted from heat from underlying volcanic flows or near surface magmas. Nevertheless a mild thermal history is suggested by: (1) the presence in the Nonesuch of abundant porphyryns (Hoering, 1965) which would probably be unstable at temperatures above 120–175° C depending on the duration of exposure to these elevated temperatures (Hoering and Abelson, 1963), (2) the presence of djurleite which strongly suggests a maximum temperature of 93° C because this sulfide breaks down to chalcocite and digenite above that temperature,³ and (3) the existence of a pink bornite which exsolves chalcopyrite when annealed at temperatures above 75° C in the same manner as other low temperature "red-bed" bornites. The lack of exsolved chalcopyrite in unannealed pink bornites suggests an origin and subsequent history at temperatures below 75° C. The reliability of this interpretation is somewhat uncertain due to an incomplete understanding of this anomalous sulfide.

While the temperature that prevailed in the Nonesuch at the time of its mineralization cannot be precisely determined, the cumulative evidence cited points to a maximum in the vicinity of 100° C. In the following calculations, a temperature of 25° C will be assumed, partly because of the availability of thermochemical data at that temperature, but also on the premise that if infiltration and/or diffusion processes can be considered effective at 25° C, they should be even more efficient at higher temperatures.

Chemical Environments of the Nonesuch and Copper Harbor Formations

It is assumed that copper was readily transported through the Copper Harbor formation and rapidly

deposited under favorable chemical conditions within the Nonesuch formation. The following analysis of the contrasting chemical environments of these two formations is an initial step in the quantitative definition of the mineralization models described here.

Nonesuch Shale.—Prior to mineralization, the Nonesuch Shale was characterized by an abundance of iron sulfide and organic matter. It is generally agreed (e.g., Berner, 1964) that stable pyrite forms in black shales from reactions involving iron with reduced sulfur generated by sulfate-reducing bacteria which thrive in organic debris. Consequently the early diagenetic environment of the Nonesuch was very probably restricted to low Eh waters.

The probable pH of the interstitial waters of the Nonesuch is not readily discernible, but a neutral to slightly acid solution appears appropriate. Bottom muds of modern basins are commonly slightly acid with the presence of dissolved H₂S and humic acid. Helgeson (1967) shows that the pH of interstitial waters in a chloritic shale may decrease markedly as temperature increases toward 300° C, but there are no indications that such extreme conditions ever existed at White Pine. In fact, the presence of disseminated copper sulfides in limy beds and calcareous concretions seems to rule out the possibility of highly acid pore solutions. The most probable Eh-pH conditions of the early Nonesuch interstitial waters are indicated by the dotted region of Figure 38.

Copper Harbor Conglomerate.—The Copper Harbor formation is a shallow-water to subaerial deposit presumably deposited under oxidizing rather than reducing conditions. Interstitial hematite is common and gives the sediment a distinctly reddish color. Sulfides and organic matter are completely lacking except in a 1–20 foot zone directly underlying the Nonesuch (Hamilton, 1967). Hamilton suspects that hematite was replaced by chlorite under reducing conditions in this local zone. There is no mineralogic evidence for such low Eh conditions in the 150–6,000 foot hematitic facies of the Copper Harbor formation proper. During mineralization, the Eh-pH conditions of the pore solutions in the Copper Harbor proper were probably not unlike that of the initial pore solutions (i.e., oxidizing and near neutral). For example, Walker et al. (1967) found diagenetic hematite forming in red-beds of the Tucson basin under oxidation potentials of about 0.500 v and pH's of about 7.8. Such an area of neutral, oxidizing conditions representative of the probable Eh-pH environment within the Copper Harbor conglomerate during mineralization is shown by hachures in Figure 38. This area lies within the stability field of Fe₂O₃ and above the stability fields of the sulfides which characterize the Nonesuch shale.

³ Discrete grains which optically resemble chalcocite and which give X-ray reflections diagnostic of djurleite could conceivably consist of chalcocite-djurleite intergrowths resultant from unmixing of high temperature hexagonal chalcocite. However, presently known phase relations of the Cu-S system (Roseboom, 1966) suggest a very limited compositional range for high temperature chalcocite, and significant amounts of djurleite could exsolve only if the initial chalcocite formed at temperatures less than about 150° C.

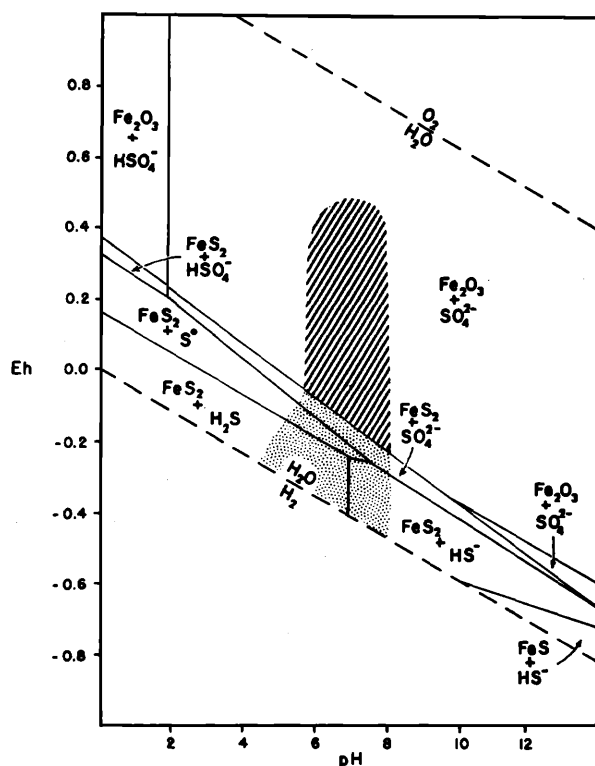


FIG. 38. Stability relations of iron sulfides, iron oxides, and sulfur species in water at 25° C and one atmosphere total pressure with total dissolved sulfur = 10^{-3} m/l (modified after Garrels and Christ, 1965). The hachured area represents the probable Eh-pH environment of interstitial water within the Copper Harbor formation, and the dotted area represents probable Eh-pH conditions of pore waters in the pyritic Nonesuch Shale.

Possible Copper Contents of Interstitial Waters

Before one can evaluate the infiltration or diffusion models, it is necessary to estimate the possible concentrations of soluble copper in the pore waters of the Copper Harbor and Nonesuch formations at the time of mineralization. In the unmineralized Nonesuch, the large amounts of sulfur available from iron sulfide would behave as a buffer on the copper concentration. At the top of the fringe, very low copper concentrations would prevail due to the formation of highly insoluble copper sulfides. High copper concentrations would be permissible within the mineralized Nonesuch (i.e., beneath the fringe) where all available sulfur had already been consumed by copper replacement of the iron sulfide. Copper could and did precipitate beneath the fringe as native copper.

The copper content of interstitial solutions in the underlying Copper Harbor formation may have been considerably greater than that of the Nonesuch Shale. Very likely the soluble copper was complexed with chloride ion which is a common constituent of

pore waters in deep portions of sedimentary basins. Samples of brine from the Keweenaw sediments at White Pine and surrounding areas commonly contain as high as 3 molar Cl^- . Using thermodynamic data from Barnes and Czamanske (1967) and Krauskopf (1967), it can be readily shown that copper is most stable as a trichloride complex in solutions containing greater than 0.6 m Cl^- . This agrees with Helgeson's theoretical studies (1968) which showed the trichloride to be more abundant than the dichloride for chloride concentrations of 1–3 molar.

Similarly it can be shown that the cupric ion will predominate over the cuprous ion only at oxidation potentials above 0.568 v. Because it is unlikely that Eh values within the natural environments of the Copper Harbor and Nonesuch formations exceeded 0.568 v, only the cuprous trichloride complex is considered below.

It can be assumed that the maximum amount of copper in the pore waters of the Copper Harbor formation would be governed by the solubility of native copper, the principal cupriferous mineral observed in this formation. The solubility of copper as CuCl_3^{2-} can then be calculated from the equi-

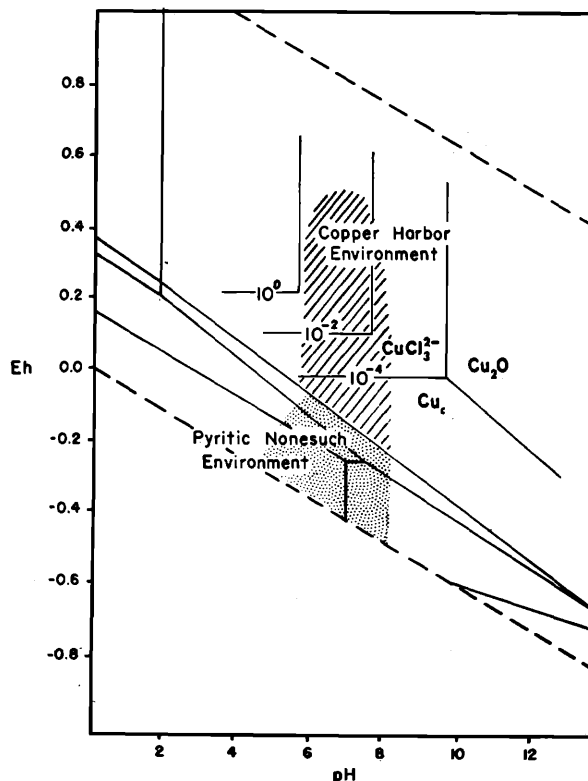
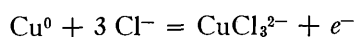


FIG. 39. Stability relations at 25° C and one atmosphere total pressure for various activities of CuCl_3^{2-} within the probable environmental limits of the Copper Harbor formation.

librium reaction:

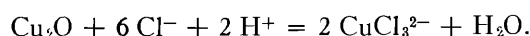


for which $\Delta F = +4.43$ kcal/mole, $E^0 = +0.192$ v, and

$$Eh = 0.192 + 0.05916 \log \frac{(\text{CuCl}_3^{2-})}{(\text{Cl}^-)^3}.$$

For an assumed chloride concentration of 3 molar and an activity coefficient, γ_{Cl^-} , of 0.7 (from Helgeson, 1964, Fig. 24), the chloride activity, (Cl^-) , is 2.1 m/l. The maximum activity of CuCl_3^{2-} can then be calculated for various portions of the Eh-pH environment of the Copper Harbor formation (see Fig. 39).

The stability limits for CuCl_3^{2-} in equilibrium with tenorite are also shown according to the equilibrium reaction:



Because tenorite is not found in the Copper Harbor conglomerate, the CuCl_3^{2-} activity was probably never greater than 10^{-2} m/l. Although the maximum activity of cuprous chloride may have been 10^{-2} m/l, the actual concentration (molarity) of CuCl_3^{2-} may have been considerably greater due to the small activity coefficient of CuCl_3^{2-} in high chloride solutions (Helgeson, 1968, Table 13). Thus it is not unlikely that the actual concentration of copper in the interstitial brine of the Copper Harbor formation could have attained values in the neighborhood of 10^{-3} m/l (64 ppm). This value, which is used in the infiltration and diffusion models below, is consistent with D. E. White's suggestion (1968, p. 324) that the ore fluid at White Pine may have "contained 100 ppm of Cu (perhaps too high,

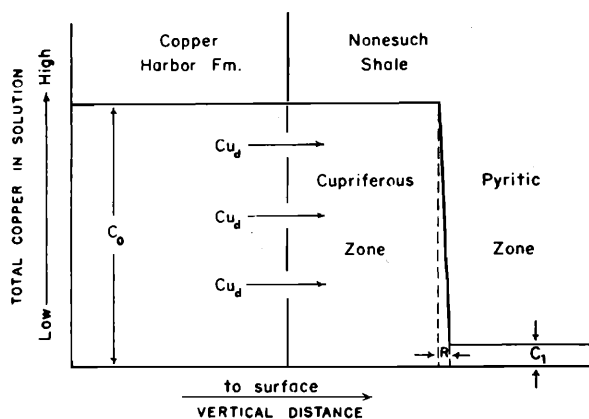


FIG. 40. Model of the infiltration process. C_0 represents the concentration of copper dissolved in interstitial solutions of the Copper Harbor formation, C_1 represents the concentration of dissolved copper due to chalcopryite in the pyritic zone, and R represents the narrow reaction zone of the Cu-Fe transition.

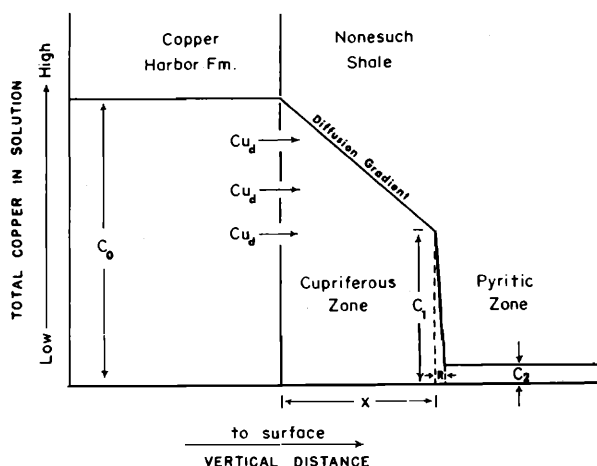


FIG. 41. Model of the diffusion process. C_0 represents the concentration of dissolved copper in interstitial solutions of the Copper Harbor formation, C_1 is the concentration of copper at the replacement front, and C_2 is the copper concentration in the pyritic zone. The distance X represents the interval between the base of the Nonesuch shale and the Cu-Fe transition, R .

but intermediate within the likely total range of 1 to 1,000 ppm of Cu)."

Infiltration Model

A strictly quantitative definition of the infiltration model (Fig. 40) is beyond reach because of our presently limited knowledge of the chemical and physical parameters involved. This situation is partially compensated by the simplicity of the mathematical operations. If A is the amount of copper entering the base of the Nonesuch shale by vertical ascension of the ore solution, then the time required to mineralize any particular bed can be calculated according to the equation:

$$T = \frac{M}{A}$$

where T is the time (sec), M is the amount of copper observed in the given bed ($\text{g} \cdot \text{cm} / \text{cm}^3$), and A is the amount of copper entering per second through each square cm of basal shale ($\text{g} / \text{cm}^2 / \text{sec}$). A is obtained from the concentration of copper in solution and the assumed vertical velocity of flow. The total duration of the mineralization period is merely the sum of the periods required to mineralize each individual bed from the base of the Nonesuch to the top of the cupriferous zone.

Table 6 shows calculated mineralization periods obtained in the above manner for several vertical drill cores at White Pine. Using an influx velocity of 10^{-8} cm/sec (~ 3 mm/yr) and a copper concentration of 10^{-3} m/l, the periods of mineralization for 15 drill cores were found to range from 312,000 to 1,190,000 years with an average of 879,000 years.

Table 6. Duration of mineralization periods estimated from infiltration and diffusion models

Drill Hole	Time (thousands of years)	
	Infiltration	Diffusion
22 $\frac{1}{2}$ D	1192	2700
23 $\frac{3}{4}$ B $\frac{1}{2}$	941	1590
25D $\frac{1}{4}$	706	1640
27 $\frac{5}{8}$ D $\frac{1}{8}$	853	2000
32D $\frac{1}{4}$	730	1440
39C	921	3570
46A	468	2650
48 $\frac{1}{2}$ C	677	2300
N28	890	1410
N37	777	3430
N47	903	1980
N48	783	1690
N67	791	1270
S8	481	1620
S9	312	800
Average	879	2010

Diffusion Model

The chemical potential necessary for the diffusion model results from estimated differences in the copper contents of interstitial solutions in the Copper Harbor (10^{-3} m/l) and the unmineralized Nonesuch formation (exceedingly small). Contributions from temperature and pressure differentials are not considered. In this ideal model, the initial concentration gradient would be infinitely steep at time zero, but would become finite as the mineralization front advanced into the overlying shale. Steady-state diffusion can be assumed after a short initial period during which copper influx may have exceeded the rate of copper deposition. With the establishment of a linear concentration gradient between the base of the shale and the mineralization front, it is possible to apply Fick's law for steady-state diffusion:

$$S = -D \frac{(\partial c)}{(\partial x)} \quad (1)$$

where S is the number of mole of copper passing per unit time across a unit surface normal to the direction of diffusion (m/sec/cm^2), D is the diffusion coefficient (cm^2/sec), c is the number of mole of copper per unit volume of solution (m/cm^3), and x is the distance of diffusion (cm). The term $\partial c/\partial x$ is the concentration gradient.

In the early stages of mineralization, the concentration gradient will be greater than in later stages because the distance x increases with time. Consequently each successively higher bed tends to require a longer period of mineralization. The rate at which the reaction front will advance is proportional to the influx of copper, S , and inversely proportional to the amount of copper, M (m/cm^3), observed in any particular bed. Thus

$$dt = \frac{M}{S} \cdot dx$$

and

$$t = \int_{x_1}^{x_2} (M/S) \cdot dx \quad (2)$$

where x_1 and x_2 are the respective distances of the bottom and top of a specific bed from the base of the Nonesuch formation. From equations (1) and (2),

$$t = - \int_{x_1}^{x_2} (M)/(D \cdot dc/dx) \cdot dx.$$

For a steady-state diffusion process, the concentration gradient, (dc/dx) , can be assumed to be $(C_0 - C_1)/x$ where C_0 and C_1 are the concentrations of copper in solution within the Copper Harbor and at the base of the mineralization front, respectively (see Fig. 41). Because of the low solubility of copper sulfides at the fringe, C_1 is much smaller than C_0 and the concentration gradient can be further approximated as C_0/x . Then

$$t = - \int_{x_1}^{x_2} (M \cdot x)/(D \cdot C_0) \cdot dx$$

or

$$t = - \frac{M(x_2^2 - x_1^2)}{2D \cdot C_0} \quad (3)$$

The precise value of the diffusion coefficient is difficult to obtain for natural systems such as copper chloride complexes migrating through the pores of a brine-saturated sediment which is composed in part of sheet silicates. However, D can probably be assumed to be within the range of 10^{-5} to 10^{-7} cm^2/sec (Garrels et al., 1949; Wehrenburg and Silverman, 1965; Peck, 1967). If the pores became severely restrictive, as may have occurred under intense compaction, the diffusibility could have decreased sufficiently to arrest the mineralization

process. For the present model, a tentative value of $D = 10^{-8}$ cm²/sec has been assumed. This value ought to be justified for at least the early history of the shale when its pore water content was large.

With this diffusion coefficient and $C_0 = 10^{-3}$ m/l, the duration of mineralization has been calculated for the same 15 drill cores of the infiltration model using equation (3). The resultant mineralization periods range from 800,000 to 3,570,000 years with an average of slightly more than 2,000,000 years (Table 6).

Discussion of Models

The estimates of the mineralization period obtained by the above infiltration and diffusion models (averages of 879,000 and 2,010,000 years, respectively) are of sufficiently short duration to suppose that mineralization of the basal Nonesuch could well have been accomplished before the permeability of the shale reached prohibitively low values under compaction. These figures certainly indicate that mineralization could have occurred before regional deformations characterized by post-ore structures such as the White Pine fault.

It is difficult to force a choice between the two models. For example, it cannot be stated with certainty that an unequivocally shorter and therefore more acceptable mineralization period results from either model, considering possible variations in the numerical values of their respective input parameters.

Other considerations lead to a similar ambiguity. The configuration of the fringe surface varies inversely with the amount of copper in the No. 21 + 23 beds and is apparently independent of the driving mechanism which brought the copper into the shale. Likewise each model is based upon an advancing reaction front producing identical paragenetic sequences of sulfides at the fringe without regard to the transport process.

One might suspect that one model offers a better explanation of the Cd and Pb anomalies which occur immediately above the fringe. The cause of these concentrations is believed to be the greater solubility of Cd and Pb sulfides relative to Cu sulfides. In the infiltration process, any traces of Cd and Pb encountered by the copper front or in the influxed ore solution itself would be swept ahead of the fringe and deposited in the presence of even more soluble iron sulfide immediately above the fringe. The diffusion process could also sweep Cd and Pb ahead of the mineralization front if a slightly decreasing concentration gradient for these metals existed in the pore solutions between the base of the Nonesuch and the fringe. No evidence of such Cd and Pb gradients need exist within the present cupriferous zone because only traces of these metals would be necessary to maintain their gradients. Again there

is no compelling reason to favor one model or the other.

A potentially valid argument for the infiltration process rather than diffusion arises from the previously described loss of sulfur which apparently resulted during replacement of pyrite by copper sulfides. An ascending interstitial solution could have readily carried away a portion of the sulfur liberated during exchange reactions at the mineralization front. Under a diffusion mechanism, however, the tendency should have been to increase sulfur within the cupriferous beds because the chemical gradient of soluble sulfur species should slope toward the region of copper sulfides whose solubilities are lower than that of pyrite. In choosing the more appropriate model, consideration should also be given to Hamilton's (1967) conclusion that downward infiltration during compaction of the shale carried soluble organic matter from the lower Nonesuch into the underlying Copper Harbor conglomerate. These concepts must be accounted for in the final analysis, but it is unlikely that present knowledge of transport processes and the succession of related geologic events is sufficient for pronouncements clearly favoring either infiltration or diffusion. The following discussions of salt-sieving and gravity stratification phenomena illustrate the need for caution in drawing conclusions at this time.

Salt Sieving Phenomena

Shales such as the Nonesuch may act as semi-permeable membranes separating more permeable sediments on either side (e.g., the Copper Harbor conglomerate below, the Freda sandstone above) and restrict the free exchange of water and dissolved constituents between the two aquifers. Because of dissolved salt, the interstitial waters of these formations can be considered as electrolytic solutions. Assuming a certain permeability, uncharged species such as water molecules may move with relative ease across a shale unit whereas charged species may be severely restricted by ion exclusion effects of charged mineral constituents such as clay particles and organic matter (DeSitter, 1947; Hanshaw, 1962; D. E. White, 1965; Berry, 1969). It is possible to enter into only a brief discussion of these interesting aspects in this paper.

During mineralization, near surface waters were presumably less saline than those at depth (Dickey, 1969). Hence there would be a tendency for dissolved salts to move upward and water molecules to move downward following the usual laws of chemical osmosis. Other processes such as electro- and thermo-osmosis and the generation of pressure differentials (Hanshaw, 1962; Hanshaw and Zen, 1965) may alter the tendency toward equilibration

of chemical potentials across the shale membrane. Furthermore, the migration of water in a shale membrane, induced by externally applied pressures or osmosis, may result in a streaming potential due to shearing of double layers composed mainly of cations associated with clay particles. As a result, net positive charges are created on low pressure surfaces of the membrane relative to high pressure regions. Ionic and molecular transfers previously discussed in the infiltration and diffusion models may be directly affected by such electric potentials across the shale. For example, the negatively-charged copper complex, CuCl_3^{2-} , would tend to be accelerated within its aqueous solvent migrating toward low pressure regions.

Osmotic and streaming potential phenomena may have operated contemporaneously with infiltration and/or diffusion processes, and may well have complemented these classic transport mechanisms in the mineralization of the lower Nonesuch Shale. A rigorous analysis of these possible effects must necessarily include simultaneous consideration of the Keweenawan basin hydrology and of evolutionary changes in basin properties with time. Such studies are now in progress (White, 1970), and will undoubtedly assist in the refinement of the basic mineralization models proposed earlier in this paper.

Gravity-Stratification Model

D. E. White (1968) has suggested, in agreement with White and Wright (1966) and the author, that the White Pine mineralization resulted from an upward migration of copper into a favorable shale host. He contends further that the upper limit of copper migration may have been defined by a gravity-stratified interface between high density brines at depth and normal surface waters above. The concept of a density-controlled interface of this type is not disputed, in fact it is supported. However, the writer (1970) disagrees with White's suggestion that the observed fringe surface at White Pine represents a former position of that interface. If instead the interface occurred well above the lower Nonesuch, one might call upon it to re-cycle the de-mineralized ore solution as described by White (p. 317). This modification is particularly desirable in the case of the infiltration model for it overcomes the serious problem of sufficient volumes of ore solution. The spent brine is simply returned to deeper portions of the basin where it may be once more mineralized and directed toward surface.

Discussion and Conclusions

Early debates over the origin of the White Pine mineralization centered around two almost overwhelming observations: (1) the widespread and re-

markably uniform distribution of copper within narrow stratigraphic limits (support for syngenesism), and (2) the close spatial association of high grade mineralization (particularly native copper) with the White Pine fault and its related anticlinal structure (support for epigenesis). A syngenetic origin is now disputed by strong evidence that copper entered the Nonesuch Shale from the underlying red-bed formation and replaced original syngenetic or early diagenetic pyrite.

Similarly, evidences such as the offset of the final position of the mineralization front (fringe) by the White Pine fault indicate that tectonic structures at White Pine are post-ore. It is still possible that portions of the high-grade sandstone copper close to the White Pine fault were introduced along this major structural break. Hamilton (1967) has observed, however, that these high grade zones are offset by minor faults which are probably related to White Pine faulting. His detailed studies of the native copper mineralization do not support the concept of fault-controlled mineralization. Ensign et al. (1968) concluded that variations between high and low grade copper in sandstone are closely related to lithologic changes and that associations of mineralization with structures such as the White Pine fault may well be fortuitous.

Evidences given in this and other papers demonstrate that most if not all copper was introduced after deposition of the lower Nonesuch Shale but before its structural deformation. The sequence of events pertaining to mineralization may be described as follows:

- 1) Deposition of the Copper Harbor conglomerate (turbulent, oxidizing conditions) followed by rather abrupt transition into Nonesuch Shale sedimentation (quiet, shallow, reducing environment). Within the lower Nonesuch there was at least one significant hiatus in sedimentation as evidenced by a minor unconformity at the top of the Parting Shale and the return to coarse-grained sandstone deposition (Upper Sandstone).

- 2) Formation of syn-diagenetic pyrite and chlorite in the early Nonesuch sediment. Bacterial reduction of sulfate in organic-rich bottom muds was probably responsible for the initial sulfide production.

- 3) An inverse relationship between the height of the fringe in the lower Nonesuch section and the amount of copper in the lowermost shale bed indicates that copper entered the shale from solutions circulating in the underlying Copper Harbor conglomerate, and that the migration of copper within the Nonesuch was essentially vertical. The ultimate source of copper was probably related directly or indirectly to Keweenawan volcanism or magmatism.

Mineralization took place along an upward advancing front (fringe) at which copper reacted with the original pyrite of the basal Nonesuch Shale in a step-by-step fashion to produce chalcocite, the principal ore mineral. A significant portion of copper and minor amounts of silver were reduced to their native elements in the lower cupriferous zone. Copper-bearing solutions may have penetrated the Nonesuch by infiltration during the early history of the shale when permeabilities and pore water contents were high. As the effective permeability was reduced by increased compaction, copper infiltration may have been succeeded by diffusion. The possible roles of salt sieving and gravity stratification processes in the transport of copper cannot be realistically evaluated at this time, but their potentials warrant serious consideration.

4) Traces of cadmium, lead (and zinc), were apparently swept upward just ahead of the copper front because of the higher solubilities of their sulfides relative to chalcocite. They form small but anomalous Cd, Pb, and (Zn) concentrations immediately above the final position of the fringe.

5) A few bedding-plane fractures and small discontinuous cross-cutting fractures permitted minor amounts of copper to migrate short distances into the pyritic zone above the final position of the fringe.

6) The main phase of copper mineralization (characterized by disseminated chalcocite and native copper with minor native silver) probably ceased when the source of metals in the underlying Copper Harbor formation was exhausted and/or when consolidation of the Nonesuch became sufficient to arrest copper migration by diffusion.

7) Moderate deformation of the Keweenaw sediments after the main phase of copper mineralization resulted in an offset of the final fringe surface along the White Pine fault. This offset is virtually identical to the offset indicated by the matching of sedimentary-facies boundaries across the fault. Local elevations of the fringe in and adjacent to fracture zones suggest that a small amount of copper was channeled upward along these open conduits. The paragenetic sequence of Cu-Fe sulfides in these "early veins" parallels that of the fringe and suggests that a close genetic relationship probably exists between the vein sulfides and sulfides within bedding. Perhaps the early vein sulfides represent an influx of the last copper-bearing solutions from the Copper Harbor formation.

8) Late vein sulfides were deposited in response to small influx of sulfur (Carpenter, 1963). This minor phase of late stage mineralization shows a paragenetic trend opposite to that represented by the sulfides of the fringe and early veins.

In the above hypothesis, deliberate distinction is

made between the origins of sulfur and copper concentrations in the lower Nonesuch shale. This is done to avoid misconceptions which might enter by undue emphasis on terms like syngenetic, epigenetic, diagenetic, and hydrothermal. For example, the entire mineralization process could conceivably be classified as diagenetic processes. The mineralizing solution may be thought of as hydrothermal, although this term could hardly mean more than warm aqueous solutions of uncertain origin in the case of the White Pine mineralization. If one were forced to give a genetic classification of the White Pine deposit on the basis of observations from this and related studies, it would probably be advisable to emphasize elemental rather than mineralogical origins: syn-diagenetic sulfur, epigenetic copper.

Acknowledgments

Sincere appreciation is expressed to C. O. Ensign, Jr. and Wm. C. Kelly who, in collaboration with W. S. White, were primarily responsible for the initiation and direction of this study. Among others who are most gratefully acknowledged are J. L. Patrick, regional geologist, J. W. Trammell, former research geologist, R. J. Leone, J. J. Fritts, D. J. Hathaway, D. H. Hopkins, R. J. Thompson, and T. L. Wright of the White Pine Copper Co. and Copper Range Co.

The project was conducted through the Department of Geology and Mineralogy, University of Michigan, under the enthusiastic supervision of Wm. C. Kelly with the additional valuable suggestions of L. I. Briggs, P. L. Cloke, C. O. Nordman, F. S. Turneaure, and the late R. M. Denning. Special thanks are due Wm. C. Kelly and P. Bartholomé, as well as the personnel of the White Pine Copper Co., for critical readings of the manuscript. The services of the Institute of Geology, University of Liege are acknowledged for aid in preparation of the manuscript. Generous financial support was received from the Copper Range Co. and a Kennecott Research Fellowship.

DEPARTMENT OF GEOLOGY AND MINERALOGY,
UNIVERSITY OF MICHIGAN,
ANN ARBOR, MICHIGAN 48104,
July 9; December 30, 1970

PRESENT ADDRESS:

ECOLE POLYTECHNIQUE,
UNIVERSITY OF MONTREAL,
2500 MARIE GUYARD,
MONTREAL 250, CANADA

REFERENCES

- Abelson, P. H., 1967, Conversion of biochemicals to kerosene and n-paraffins in *Researches in Geochemistry*, ed. P. H. Abelson: John Wiley and Sons, Inc., p. 63-86.
- Adler, I., 1966, *X-ray Emission Spectrography in Geology*: Elsevier Publishing Co., New York, 258 pp.

- Baas, Becking, L. G. M., Kaplan, I. R., and Moore, D., 1960, Limits of the natural environment in terms of pH and oxidation-reduction potentials: *J. Geol.*, v. 68, p. 243-284.
- Baas Becking, L. G. M., and Moore, D., 1961, Biogenic sulfides: *ECON. GEOL.*, v. 56, p. 259-272.
- Barghoorn, E. S., Meinschein, W. G., and Schopf, J. W., 1965, Paleobiology of a Precambrian shale: *Science*, v. 148, p. 461-472.
- Barnes, H. L., and Czamanske, G. K., 1967, Solubilities and transport of ore minerals in *Geochemistry of Hydrothermal Ore Deposits*, ed. H. L. Barnes: Holt, Rinehart and Winston, Inc., p. 334-381.
- Barton, P. B., Jr., and Skinner, B. J., 1967, Sulfide mineral stabilities in *Geochemistry of Hydrothermal Ore Deposits*, ed. H. L. Barnes: Holt, Rinehart and Winston, Inc., p. 236-333.
- Berner, R. A., 1964, Distribution and diagenesis of sulfur in some sediments from the Gulf of California: *Marine Geol.*, v. 1, p. 117-140.
- Berry, F. A. F., 1969, Relative factors influencing membrane filtration effects in geologic environments: *Chem. Geol.*, v. 4, p. 295-301.
- Berry, L. G., and Thompson, R. M., 1962, X-ray powder data for ore minerals: *The Peacock Atlas*, *Geol. Soc. Amer. Mem.* 85, 281 pp.
- Birks, L. S., 1959, *X-ray Spectrochemical Analysis*: Interscience Publishers, Inc., New York, 137 pp.
- Brecke, E. A., 1968, Copper mineralization in the upper part of the Copper Harbor Conglomerate at White Pine (discussion): *ECON. GEOL.*, v. 63, p. 294.
- Brett, R., and Yund, R. A., 1964, Sulfur-rich bornites: *Am. Mineralogist*, v. 49, p. 1084-1098.
- Brown, A. C., 1965, Mineralogy at the top of the cupriferous zone, White Pine mine, Ontonagon Co., Michigan: unpublished M.S. thesis, The University of Michigan, Ann Arbor, 81 pp.
- , 1968, Zoning in the White Pine copper deposit, Ontonagon County, Michigan: Doctoral dissertation, University of Michigan, University Microfilms, Inc., No. 69-12,057, Ann Arbor, 199 pp.
- , 1970, Environments of generation of some base-metal ore deposits (discussion): *ECON. GEOL.*, v. 65, p. 60-61.
- , and Trammell, J. W., 1966, Zoning of the White Pine copper deposit, Ontonagon Co., Michigan: *ECON. GEOL.*, v. 61, p. 1308, Abstract.
- Butler, B. S., and Burbank, W. S., 1929, The copper deposits of Michigan: U. S. Geol. Survey Prof. Paper 144.
- Carpenter, R. H., 1963, Some vein-wall rock relationships in the White Pine mine, Ontonagon Co., Michigan: *ECON. GEOL.*, v. 58, p. 643-666.
- Chaudhuri, S., and Faure, G., 1967, Geochronology of the Keweenaw rocks, White Pine, Michigan: *ECON. GEOL.*, v. 62, p. 1011-1033.
- Davidson, C. F., 1962, The origin of some strata-bound sulfide ore deposits: *ECON. GEOL.*, v. 57, p. 265-275.
- Degens, E. T., 1965, *Geochemistry of Sediments*: Prentice-Hall, Inc., Englewood Cliffs, New Jersey, 342 pp.
- DeSitter, L. U., 1947, Diagenesis of oil-field brines: *Am. Assoc. Petroleum Geologists Bull.*, v. 31, p. 2030-2040.
- Dickey, P. A., 1969, Increasing concentration of subsurface brines with depth: *Chem. Geol.*, v. 4, p. 361-370.
- Ensign, C. O., Jr., White, W. S., Wright, J. C., Patrick, J. L., Leone, R. J., Hathaway, D. J., Trammell, J. W., Fritts, J. J., and Wright, T. L., 1968, Copper deposits in the Nonesuch Shale, White Pine, Michigan in *Ore Deposits of the United States 1933-1967*, ed. J. D. Ridge: Am. Inst. Mine. Metal. and Petr. Eng., Inc., p. 460-488.
- Garlick, W. G., 1961, The syngenetic theory in *The Geology of the Northern Rhodesian Copperbelt*, ed. F. Mendelson: MacDonald and Co., Ltd., London, 523 p.
- Garrels, R. M., and Christ, C. L., 1965, *Solutions, Minerals, and Equilibria*: Harper and Row Publishers, Inc., New York, 450 pp.
- Garrels, R. M., Dreyer, R. M., and Howland, A. L., 1949, Diffusion of ions through intergranular spaces in water-saturated rocks: *Bull. Geol. Soc. Am.*, v. 60, p. 1809-1828.
- Hamilton, S. K., 1967, Copper mineralization in the upper part of the Copper Harbor conglomerate at White Pine, Michigan: *ECON. GEOL.*, v. 62, p. 885-904.
- Hanshaw, B. B., 1962, Membrane properties of compacted clays: unpublished Ph.D. thesis, Harvard University, 113 pp.
- , and Zen, E-an, 1965, Osmotic equilibrium and overthrust faulting: *Geol. Soc. Amer. Bull.*, v. 76, p. 1376-1386.
- Helgeson, H. C., 1964, *Complexing and Hydrothermal Ore Deposition*: The MacMillan Co., New York, 128 pp.
- , 1967, Silicate metamorphism in sediments and the genesis of hydrothermal solutions: *ECON. GEOL.*, Monograph 3, p. 33-342.
- , 1968, Thermodynamics of hydrothermal systems at elevated temperatures and pressures: *Am. Jour. Sci.*, v. 267, p. 729-804.
- Hite, D. M., 1968, Sedimentology of the Upper Keweenaw sequence of northern Wisconsin and adjacent Michigan: unpublished Ph.D. thesis, University of Wisconsin, 204 pp.
- Hoering, T. C., 1965, The extractable organic matter in Precambrian rocks and the problem of contamination in *Annual Report of the Director, Geophysical Laboratory*: Carnegie Institution Year Book 64, 250 pp.
- , and Abelson, P. H., 1963, Hydrocarbons from kerosene in *Annual Report of the Director, Geophysical Laboratory*: Carnegie Institution Yearbook 62, p. 229-234.
- Kelly, W. C., 1962, Unpublished map files of the White Pine Copper Co.
- Krauskopf, K., 1967, *Introduction to Geochemistry*: McGraw-Hill Books Co., New York, 721 pp.
- Leith, C. K., Lund, R. J., and Leith, A., 1935, Precambrian rocks of the Lake Superior region: U. S. Geol. Surv. Prof. Paper 184, 34 pp.
- Liebhafsky, H. A., Pfeiffer, H. C., Winslow, F. H., and Zemany, D. D., 1960, X-ray absorption and emission in analytical chemistry: John Wiley, New York, 357 pp.
- Love, L. G., 1962, Biogenic primary sulfides of the Permian Kupferschiefer and Marl slate: *ECON. GEOL.*, v. 57, p. 250-366.
- , and Zimmerman, D. O., 1961, Bedded pyrite and microorganisms from the Mount Isa shale: *ECON. GEOL.*, v. 56, p. 873-896.
- Ohle, E. L., 1968, Copper mineralization in the upper part of the Copper Harbor conglomerate at White Pine, Michigan: *ECON. GEOL.*, v. 63, p. 190-191.
- Parravano, C., Baldeschwieler, J. D., and Boudart, M., 1967, Diffusion of water in zeolites: *Science*, v. 155, p. 1535-1536.
- Peck, A. J., 1967, Mass transport in porous rocks: *Mineralium Deposita*, v. 2, p. 243-247.
- Roseboom, E. H., Jr., 1962, Djurleite, $\text{Cu}_{1.66}\text{S}$, a new mineral: *Am. Mineralogist*, v. 47, p. 1181-1184.
- , 1966, An investigation of the system Cu-S and some natural copper sulfides between 25° C and 700° C: *ECON. GEOL.*, v. 61, p. 641-672.
- Russell, R. D., and Farquhar, R. M., 1962, Rapid analysis of silicate, carbonate, and phosphate rocks: U. S. Geol. Surv. Bull. 1144-A, 56 pp.
- Skinner, B. J., 1966, The system Cu-Ag-S: *ECON. GEOL.*, v. 61, p. 1-26.
- Smith, F. G., 1955, Structure of zinc sulfide minerals: *Am. Mineralogist*, v. 40, p. 658-675.
- Temple, K. L., and LeRoux, N. W., 1964, Syngensis of sulfide ores: Sulfate-reducing bacteria and copper toxicity: *ECON. GEOL.*, v. 59, p. 271-278.
- Trammell, J. W., and Ensign, C. O., Jr., 1964, Ancient stream channels and their effect on mine planning and grade control at the White Pine Mine, Michigan: Society of Mining Engineers of AIME, Fall meeting, New York.
- U. S. Geological Survey, 1966, Epigenetic copper mineralization in Nonesuch shale of Michigan: U. S. Geol. Surv. Prof. Paper 550-A, p. A3.
- Van Hise, C. R., and Leith, C. K., 1911, The geology of the Lake Superior region: U. S. Geol. Surv., Mon. 52, p. 411.

- Walker, T., Back, W., and Laney, R. L., 1967, Oxidation potentials in water: U. S. Geol. Surv. Prof. Paper 575-A, p. A162.
- Wedepohl, L. H., 1964, Untersuchungen am Kupferschiefer in Nordwestdeutschland; Ein Beitrag zur Deutung der Genese bituminöser Sedimente: *Geochim. et Cosmochim. Acta*, v. 28, p. 305-364.
- Wehrenberg, J. P., and Silverman, A., 1965, Studies of base metal diffusion in experimental and natural systems: *ECON. GEOL.*, v. 60, p. 317-350.
- White, D. E., 1965, Saline waters of sedimentary rocks in Fluids in Subsurface Environments: *Am. Assoc. Petroleum Geologists Memoir* 4, p. 342-366.
- , 1968, Environments of generation of some base-metal ore deposits: *ECON. GEOL.*, v. 63, p. 301-335.
- White, W. S., 1960, The White Pine copper deposit (discussion): *ECON. GEOL.*, v. 55, p. 401-414.
- , 1970, A paleohydrologic model for mineralization of the White Pine copper deposit, northern Michigan: *Geol. Soc. Amer. Annual Meeting 1970—Abstracts*, p. 721.
- , and Wright, J. C., 1954, The White Pine copper deposit, Ontonagon County Michigan: *ECON. GEOL.*, v. 49, p. 675-726.
- , and —, 1960, Lithofacies of the Copper Harbor conglomerate, northern Michigan: U. S. Geol. Surv. Prof. Paper 400-B, p. B5-8.
- , and —, 1966, Sulfide-mineral zoning in the basal Nonesuch shale, northern Michigan: *ECON. GEOL.*, v. 61, p. 1171-1190.
- Wiese, R. G., Jr., 1960, Petrology and geochemistry of a copper-bearing Precambrian shale, White Pine, Michigan: unpublished Ph.D. thesis, Harvard University, Cambridge.
- Yund, R. A., and Kullerud, G., 1966, Thermal stability of assemblages in the Cu-Fe-S system: *Jour. Petr.*, v. 7, p. 454-488.